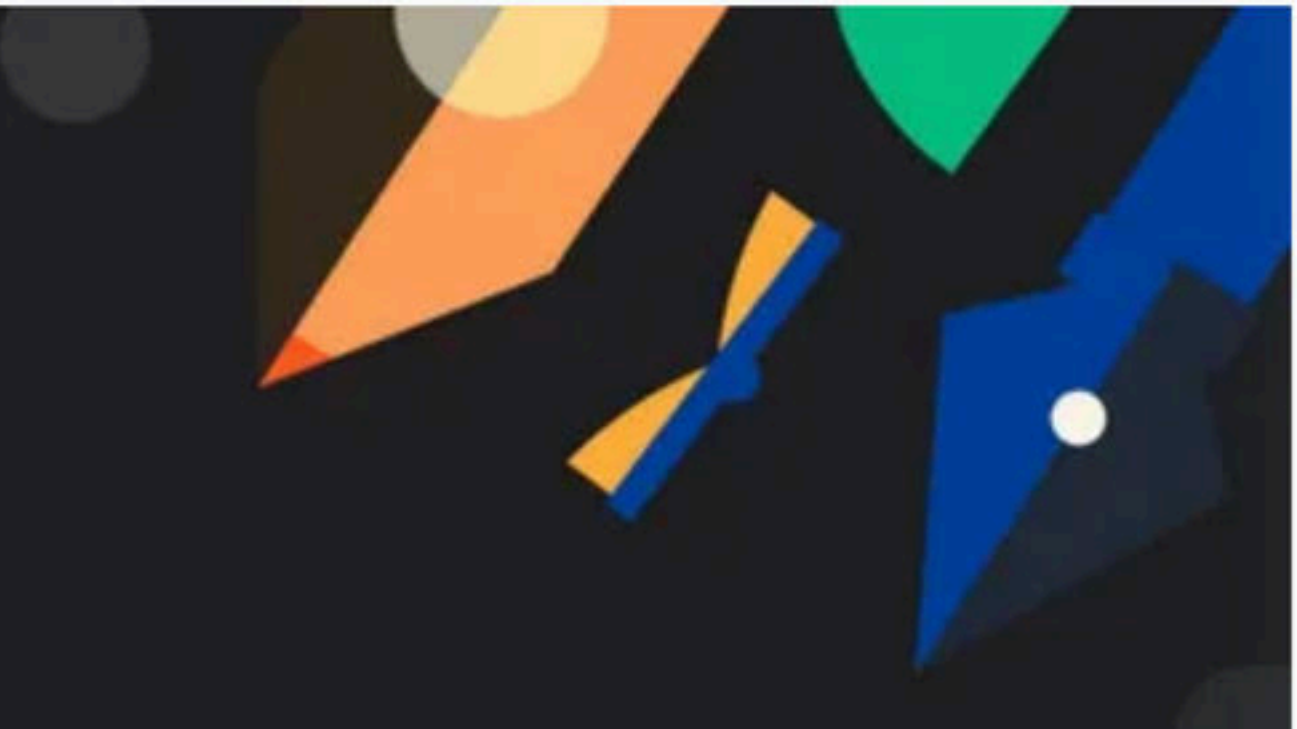


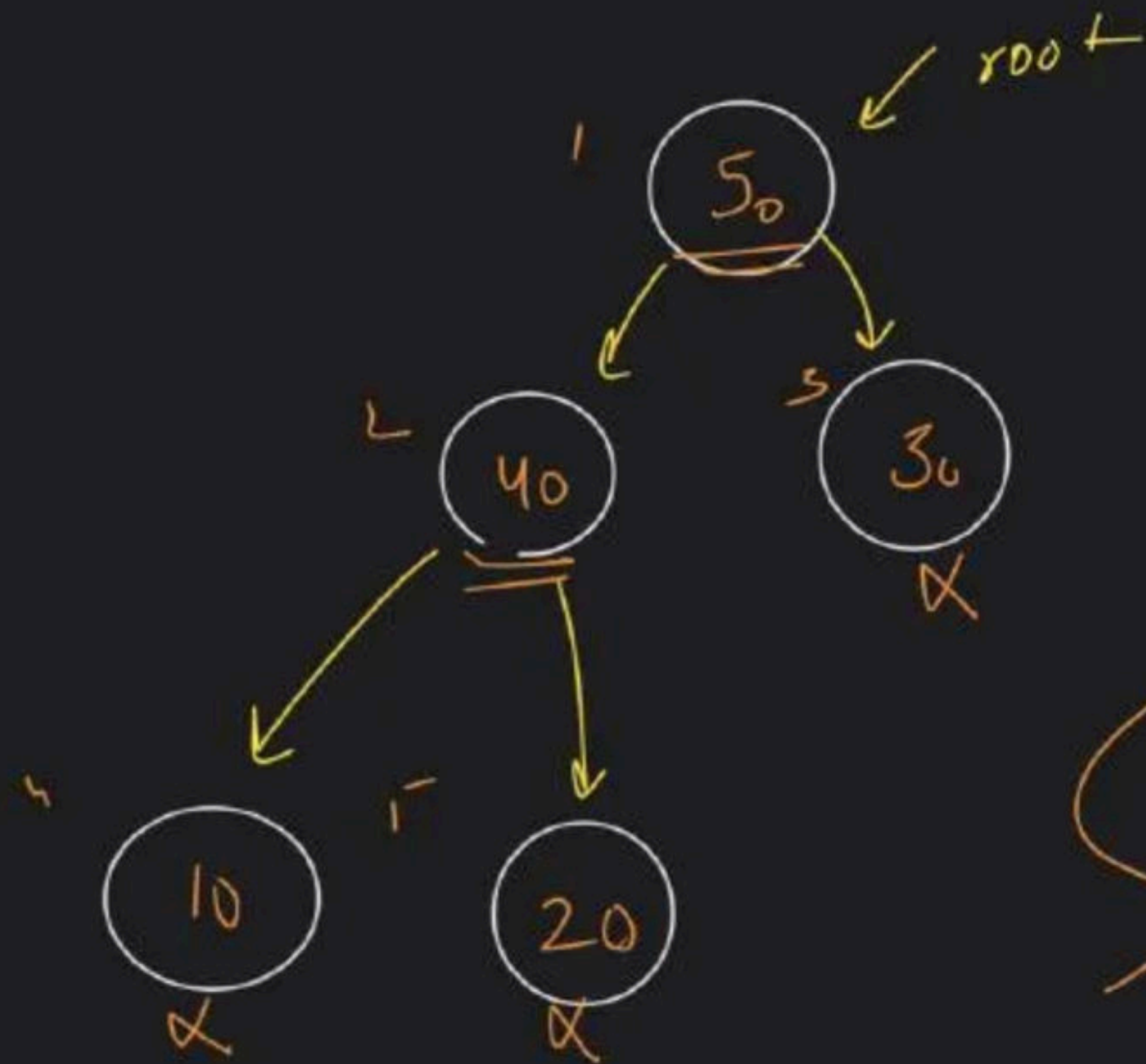


Doubt Class with Lakshay

Special class

Heaps Class - 2

Special class



50 40 30 10 20

Heap

?

D.S

C.B.T + Heap Property Satisfy

Min / Max
parent / root
↓
parent / root
↓
> l, > r

C.B.T

①

except last level - α

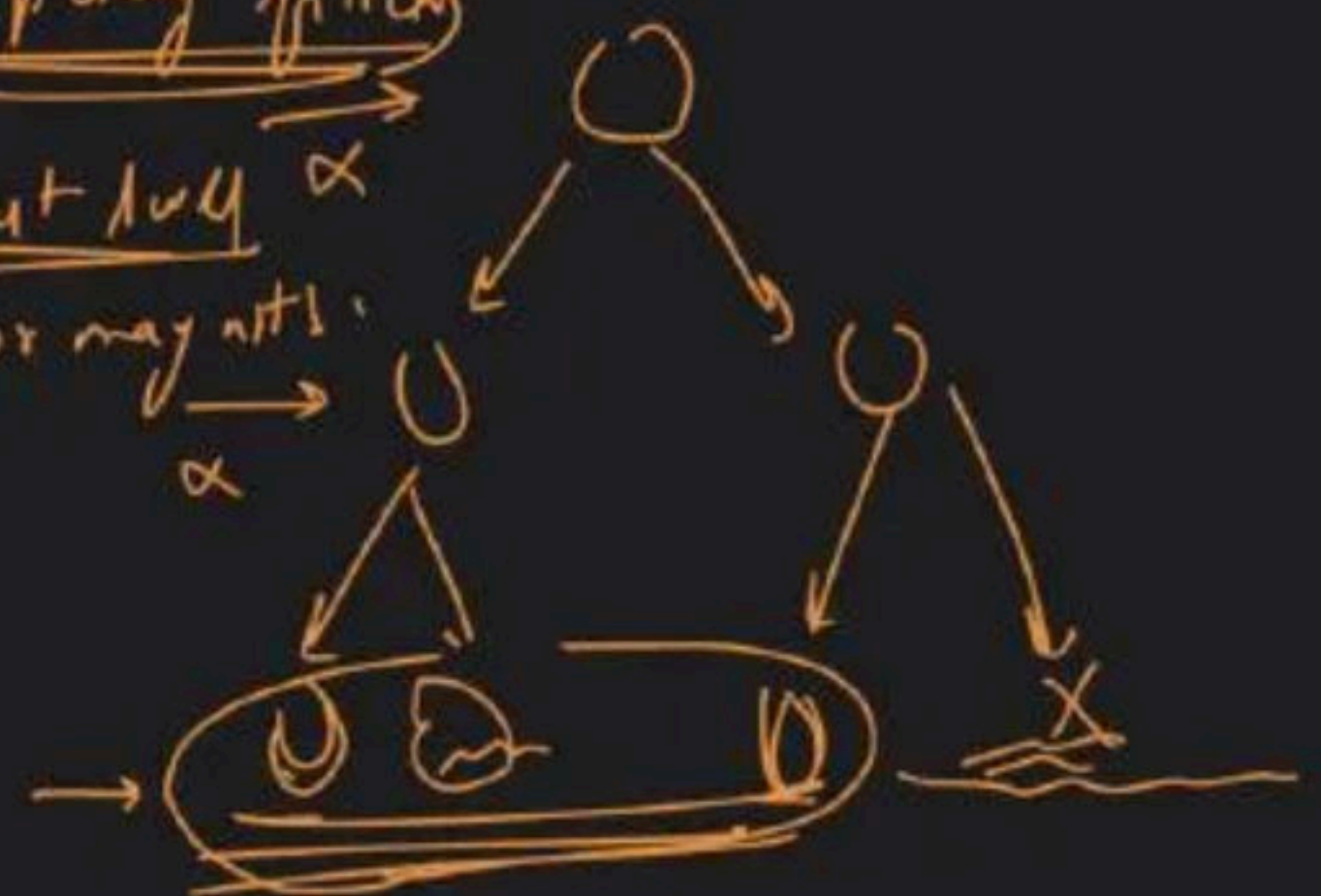
all levels are completely filled

~~nodes~~

except the last level α
may or may not be

② nodes fill

left → right [continuous manner]



$\circ \times \times \circ \times$

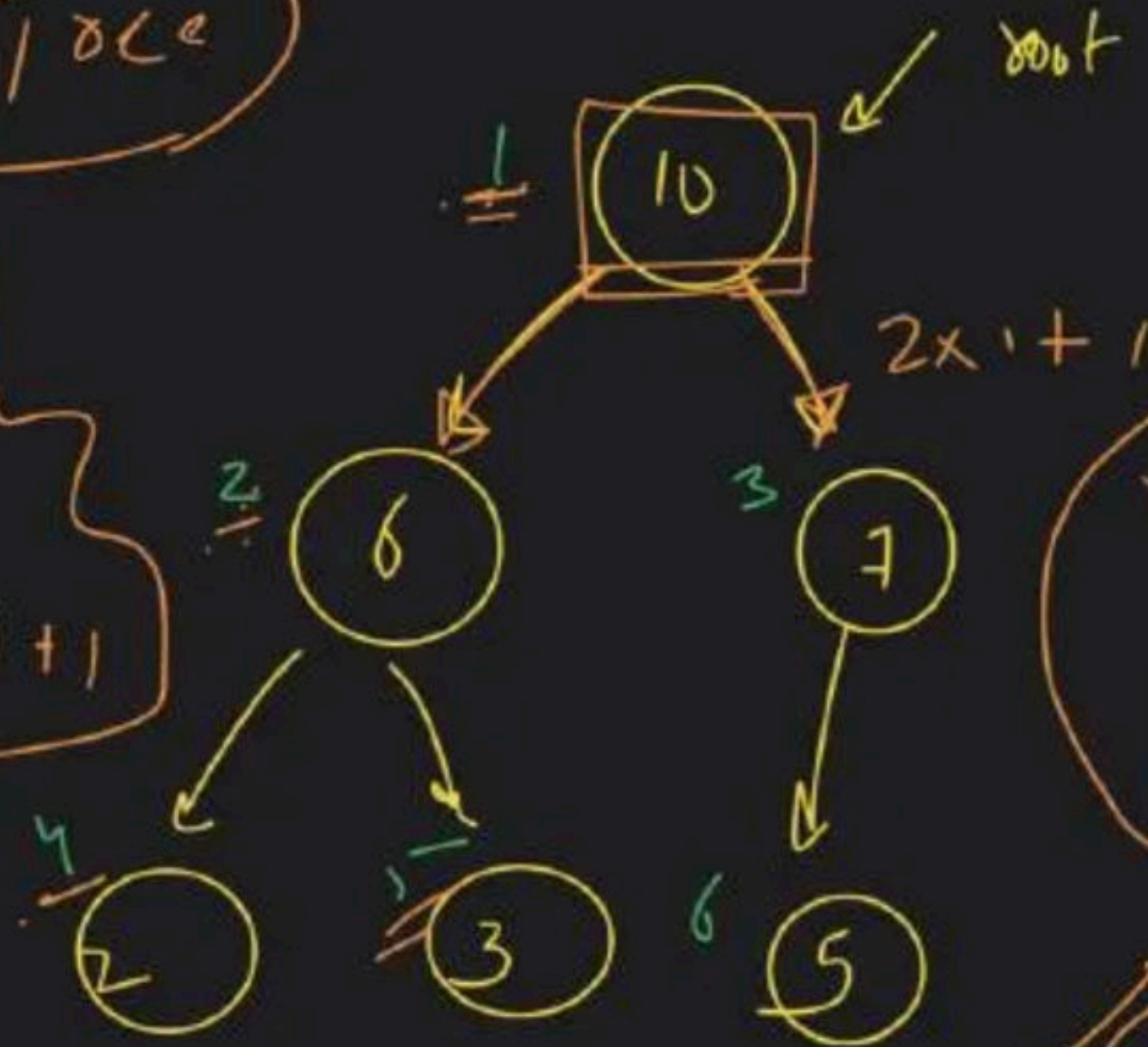
$\circ \circ \circ \circ$

→ Heap → visualization → Tree

Impl

array

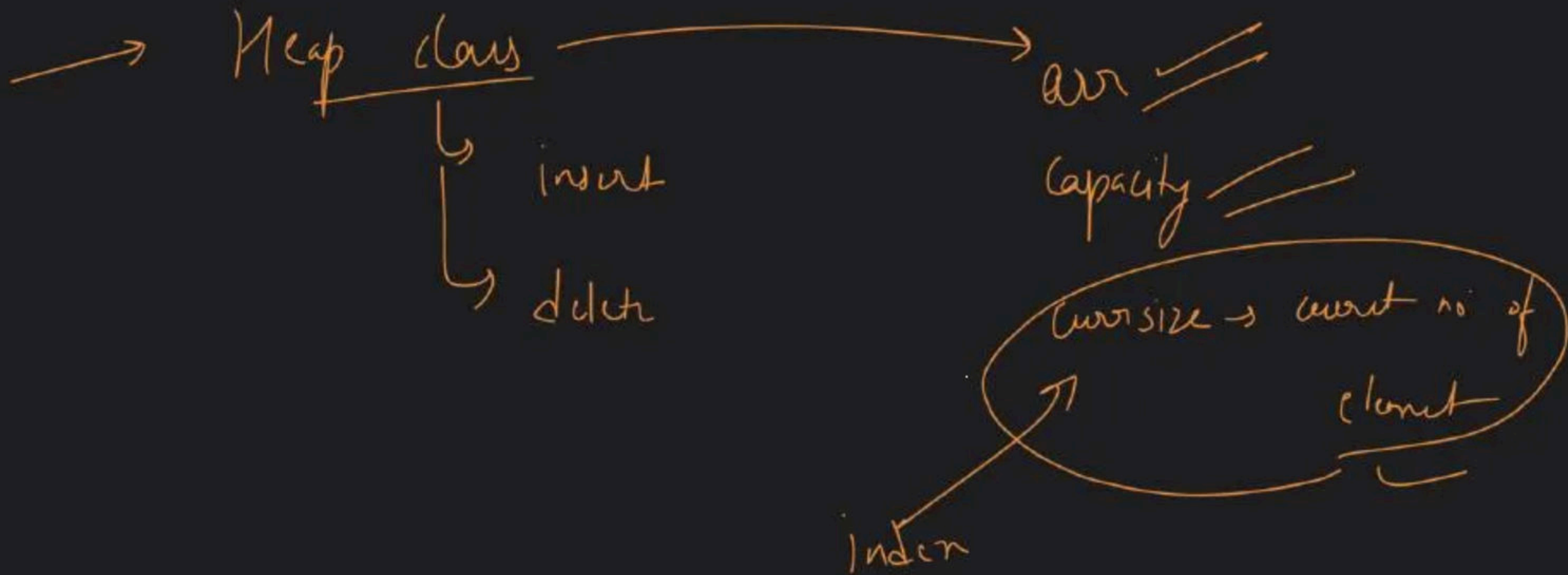
parent = i
left child = $2 \star i$
right child = $2 \star i + 1$



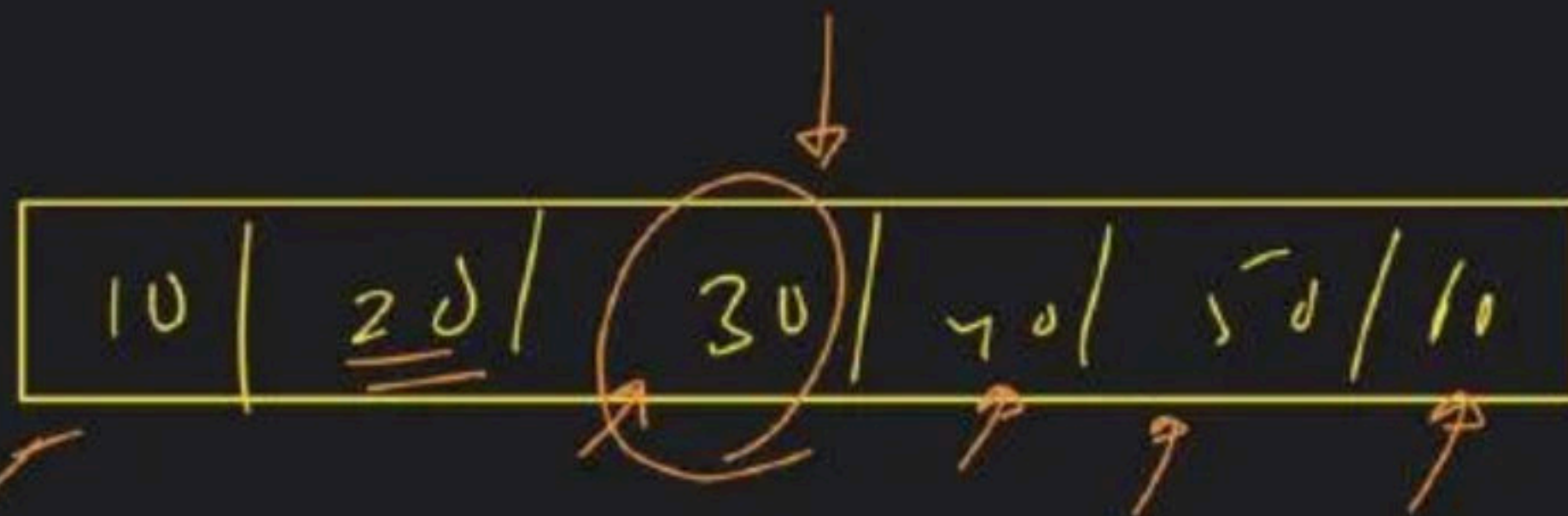
right child = i
↓
parent = $i/2$

left child = i
↓
parent = $i/2$

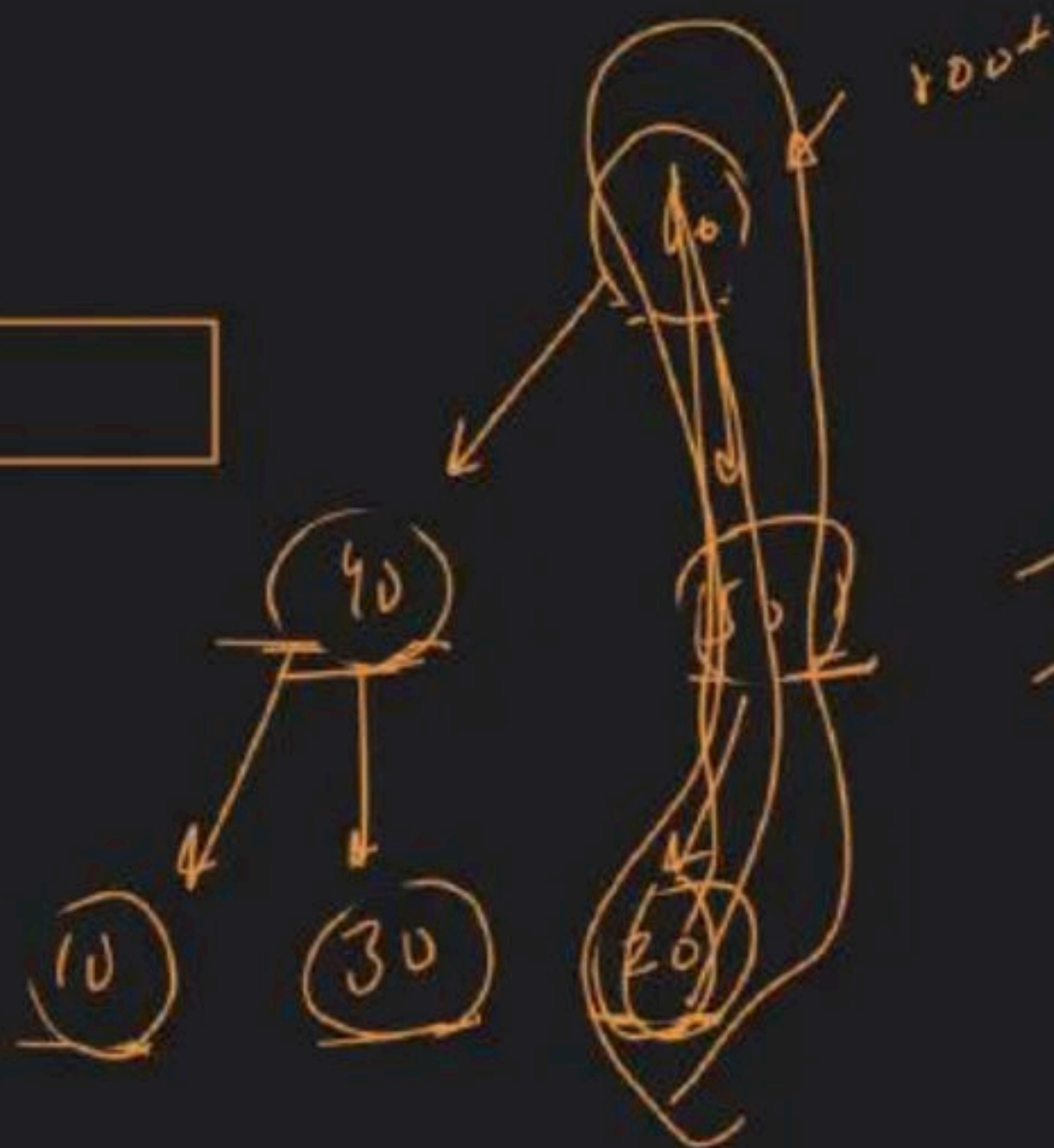
X	10	6	7	2	3	5
0	1	2	3	4	5	6



→ insert
 ↳ $O(\log n)$



~~iden~~



(A) insert → Level order wise

(B) node → right position
 ↓
 by comparing with parent

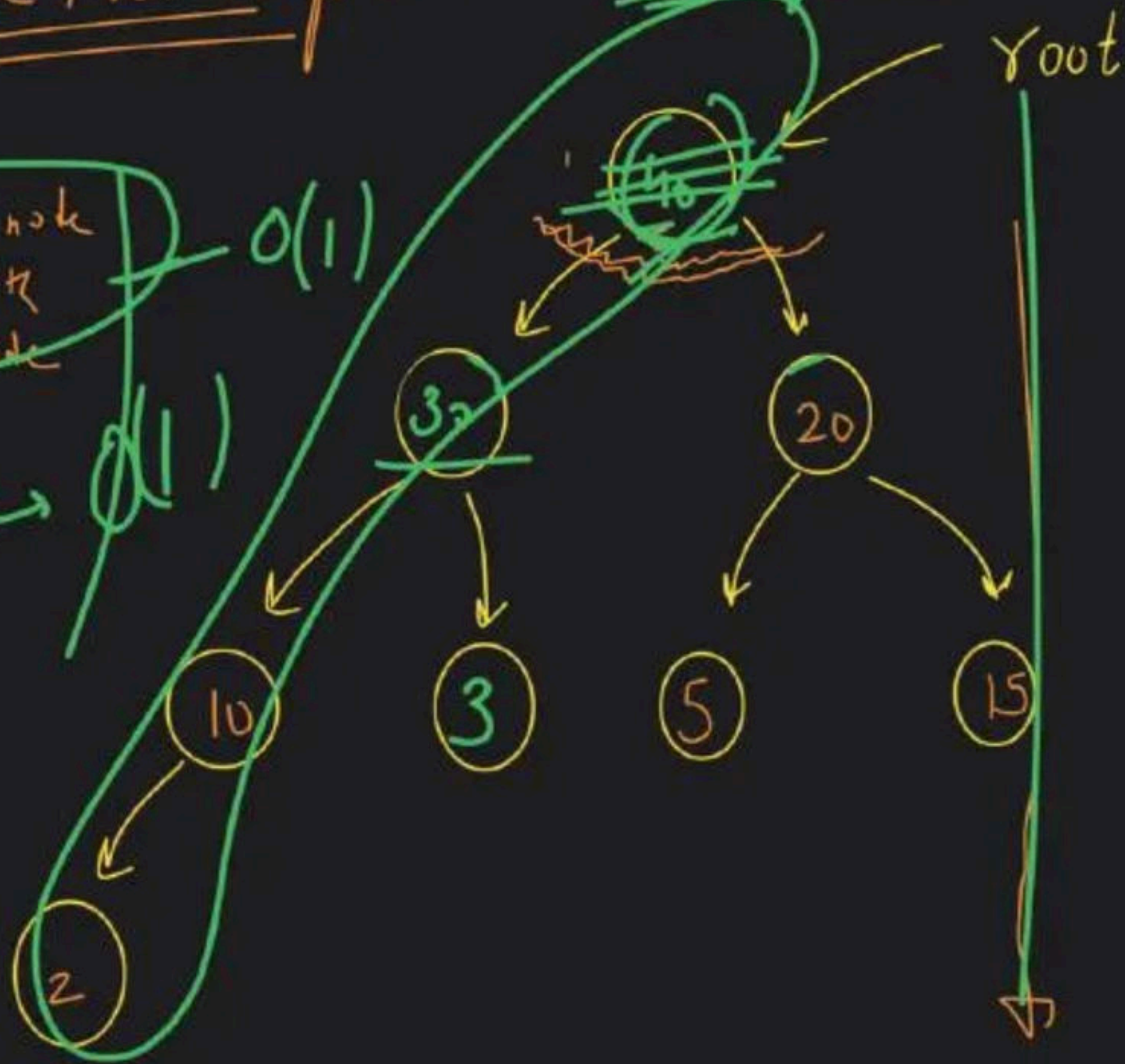
→ delete from Heap → $O(\log n)$

① replace root node with last node $O(1)$

② $n \rightarrow$ index → $O(1)$

③ arr[i] → val

right position
↓
heapify



heapify

↓
index

left / right
val

largest Val, index

↓ ↓
L R

swap (in, largest)

// Back to recur.

→ insert $\rightarrow O(\log n)$

arr $\rightarrow$ n element



in-place

$O(n)$



Build Map

$\rightarrow$ $\frac{n \log n}{2}$

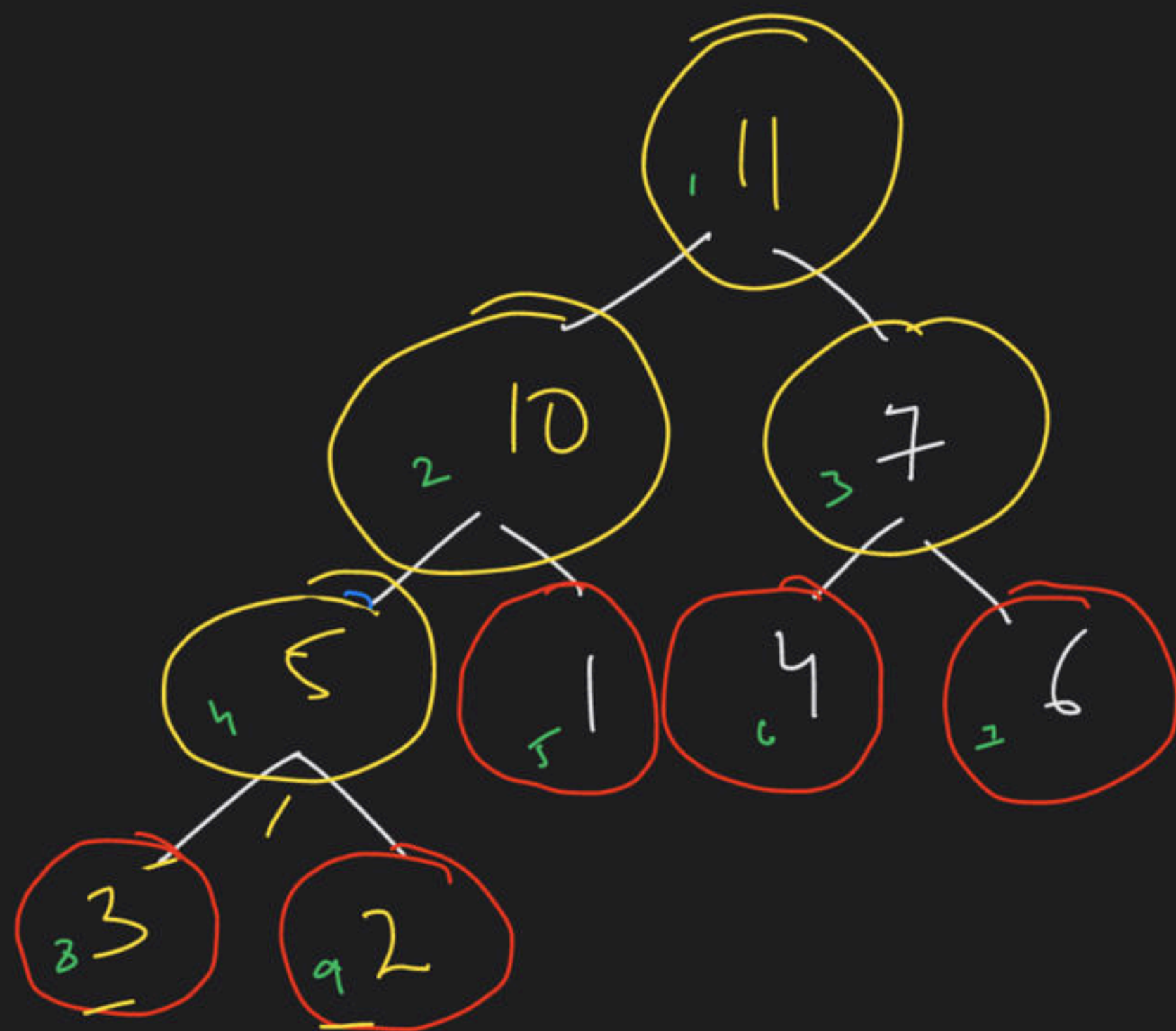
Max Heap

==

10	3	5	7	2	1	4	6	10	11
8	1	2	3	4	5	6	7	8	9

$$N = 9, \quad N/2 \Rightarrow 4$$

(N)



Swap

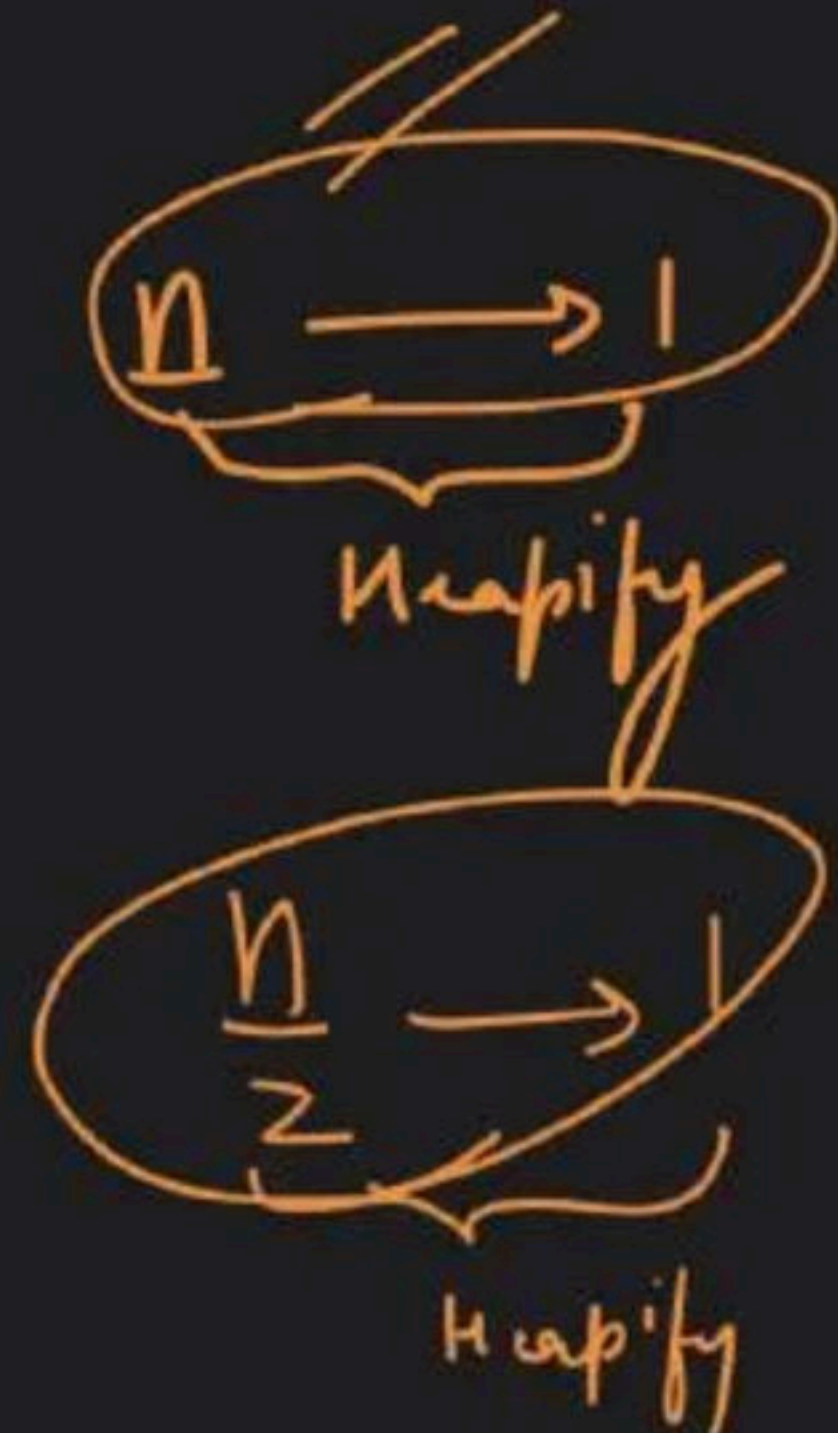
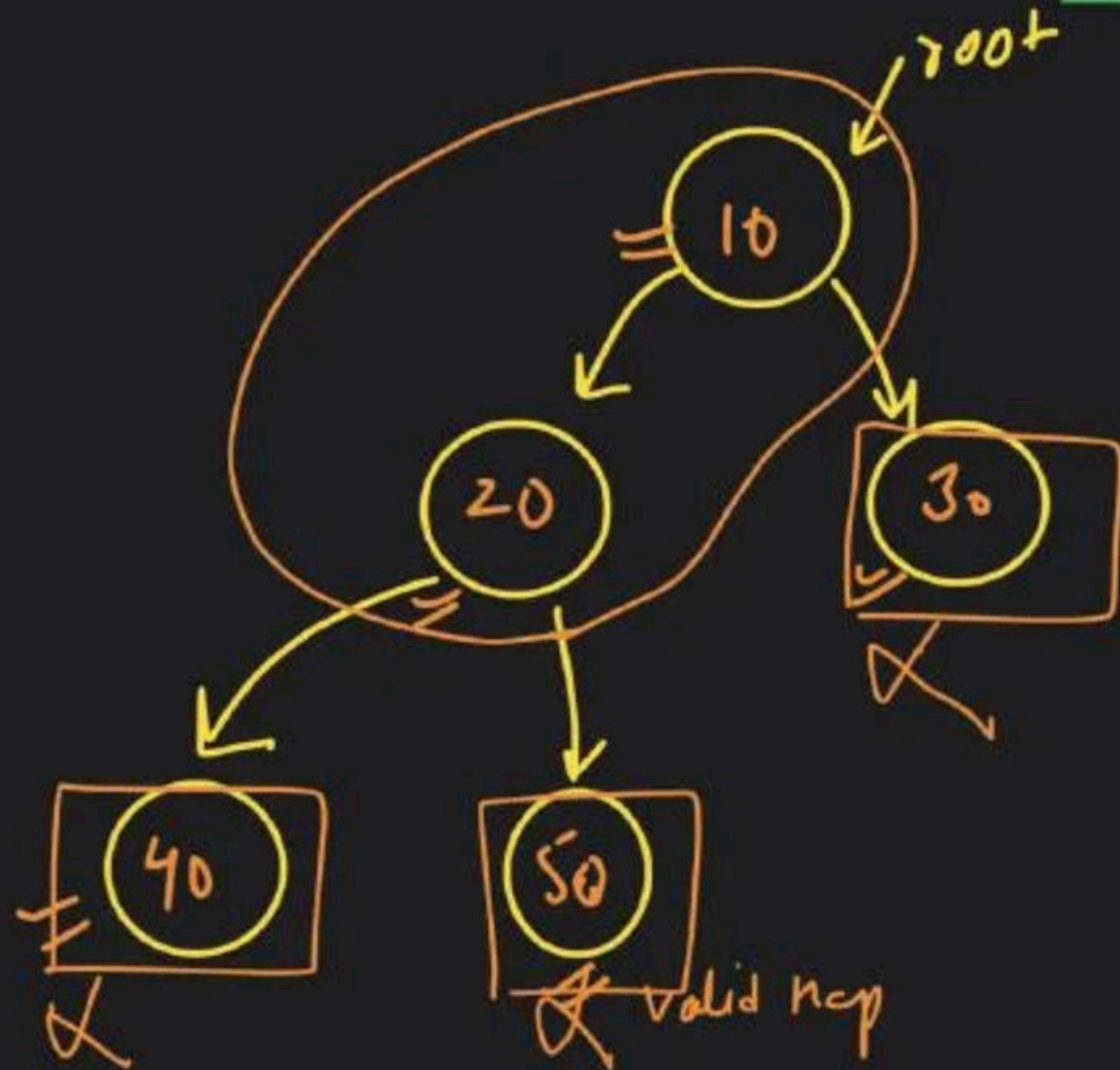
2 ↔ 4

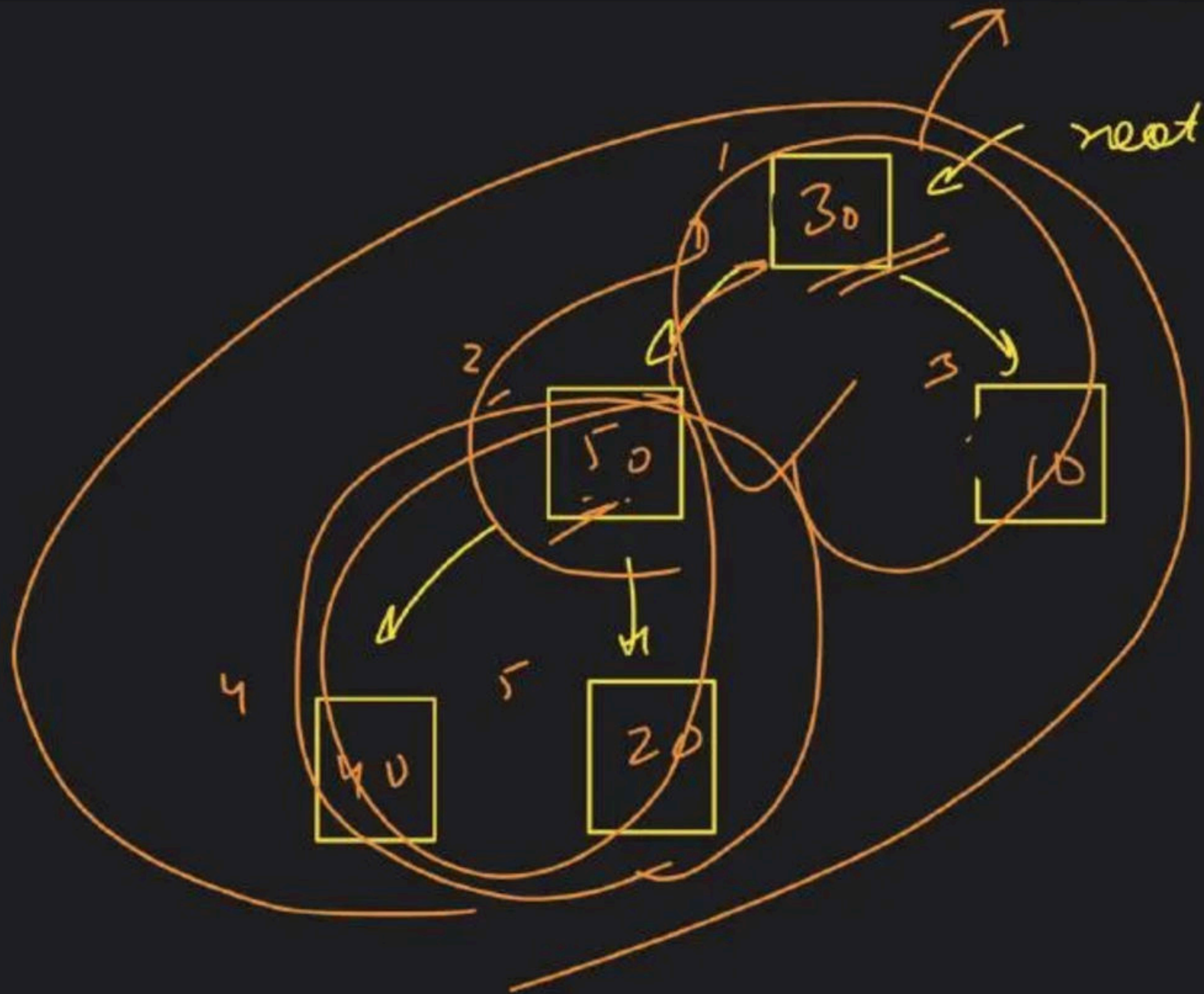
4 ↔ 8



$$\left(\frac{N}{2} \times \log N \right)$$

1	2	3	4	5
10	20	30	40	50





$$\checkmark \frac{n}{2} \rightarrow 1$$

or



$$1 \rightarrow \frac{5}{2}$$

$$1 \rightarrow 2$$

Heap Sort :-

1 2 3 4 5
~~50~~, 40, 30, 10, 20
20

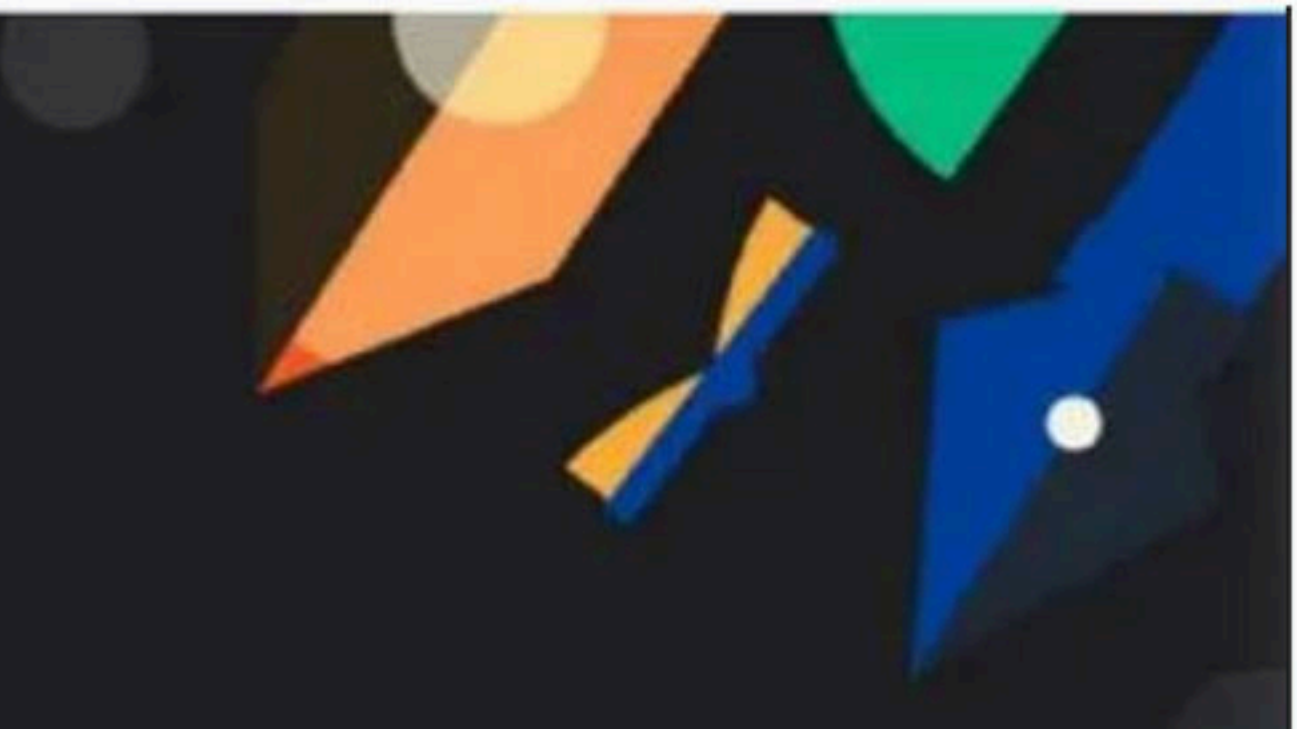
(n-1) baar
delux op

~~40~~, 20, 30, 10 } 50
10

after
heapify → ~~30~~, 20, 10 } 40, 50
10

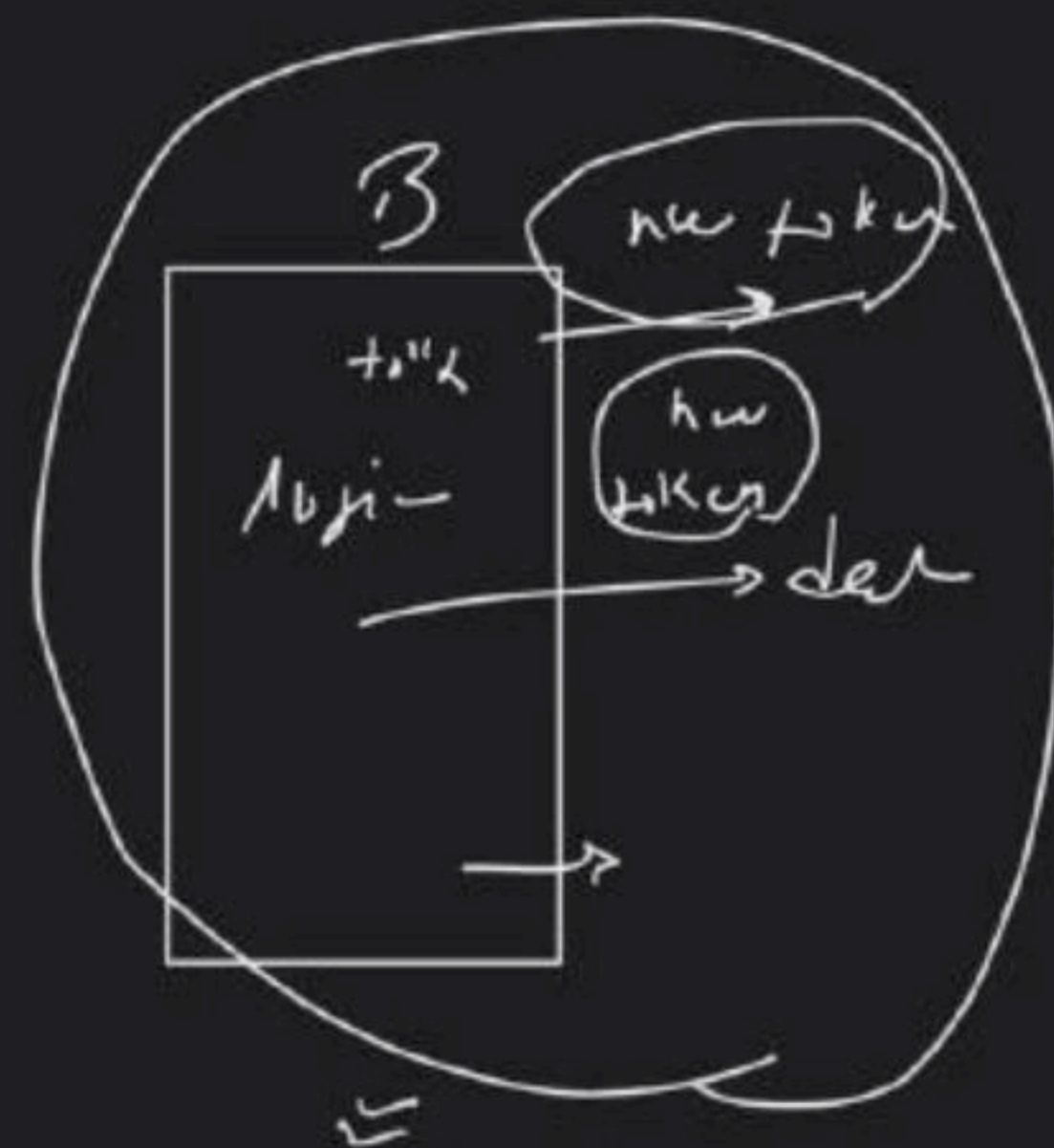
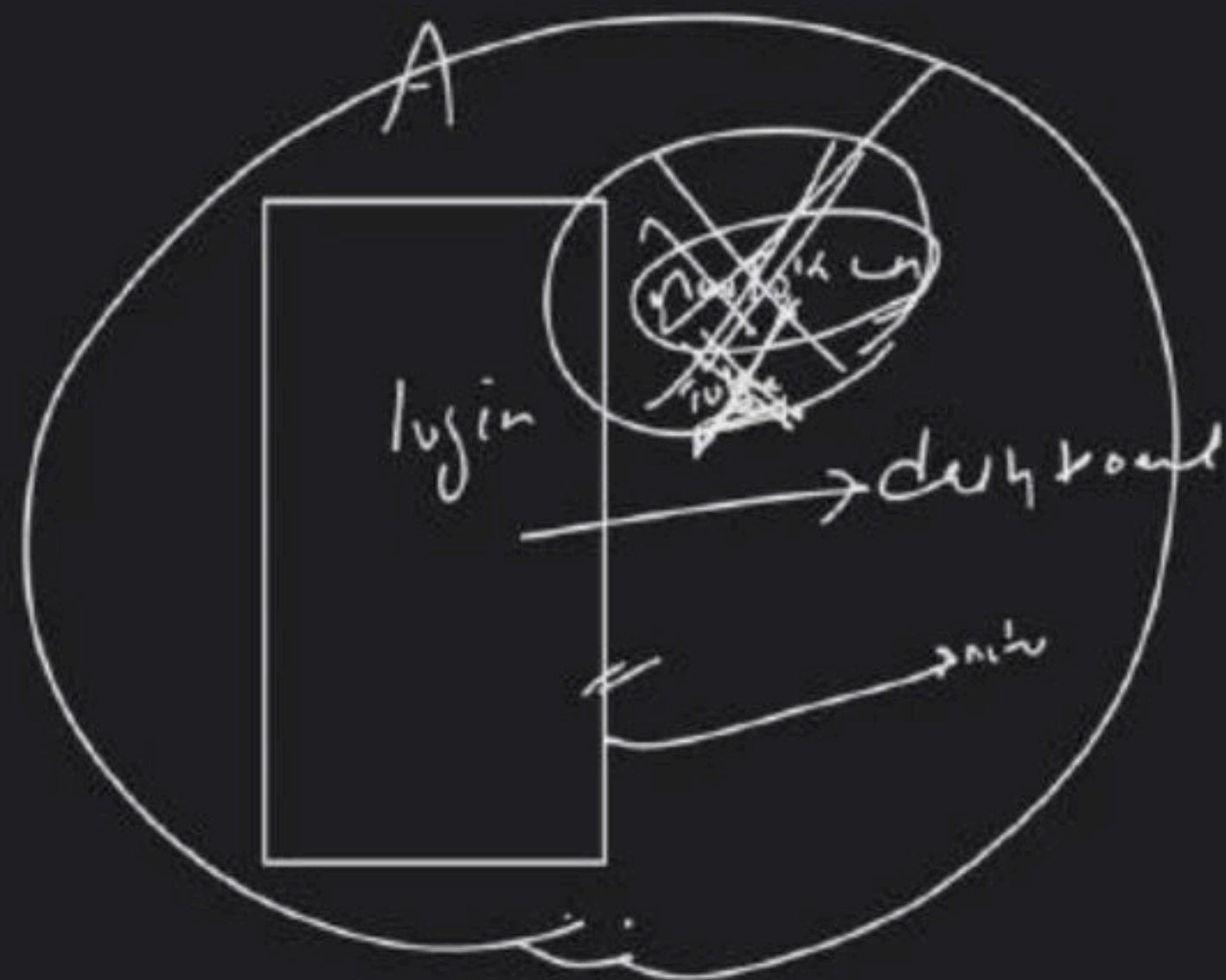
~~20~~, 10 } 30, 40, 50
10

10 } 20, 30, 40, 50 → Sort?

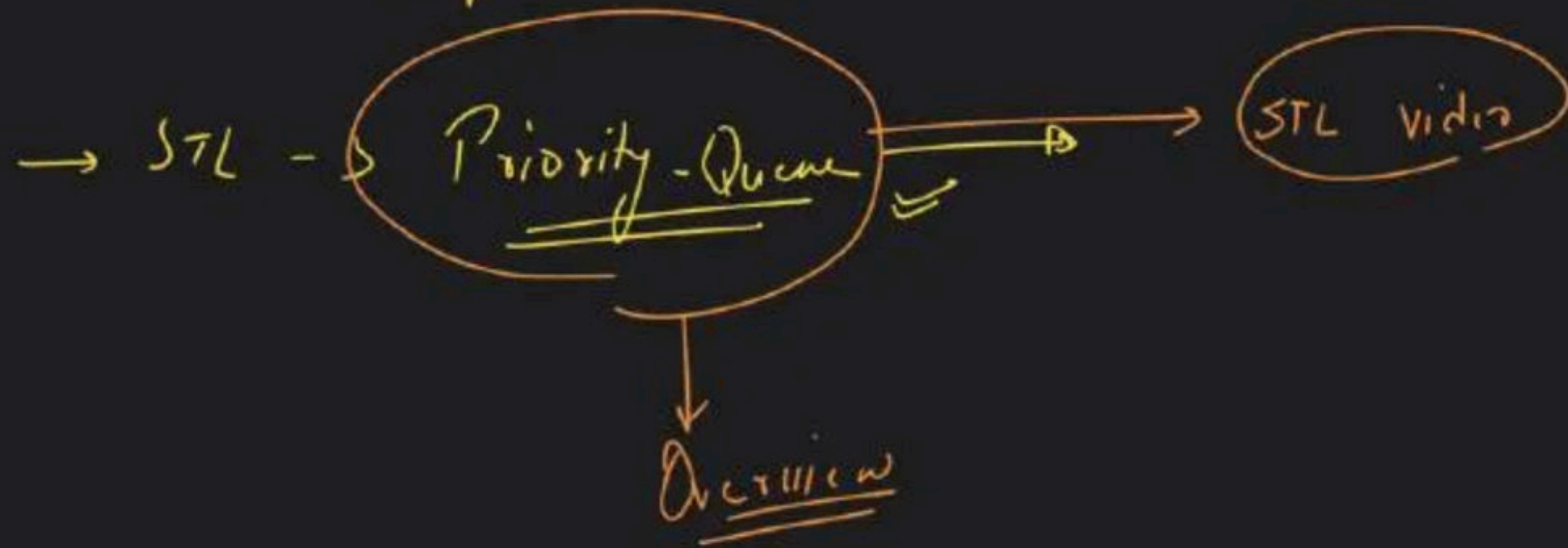


Heaps Class - 3

Special class

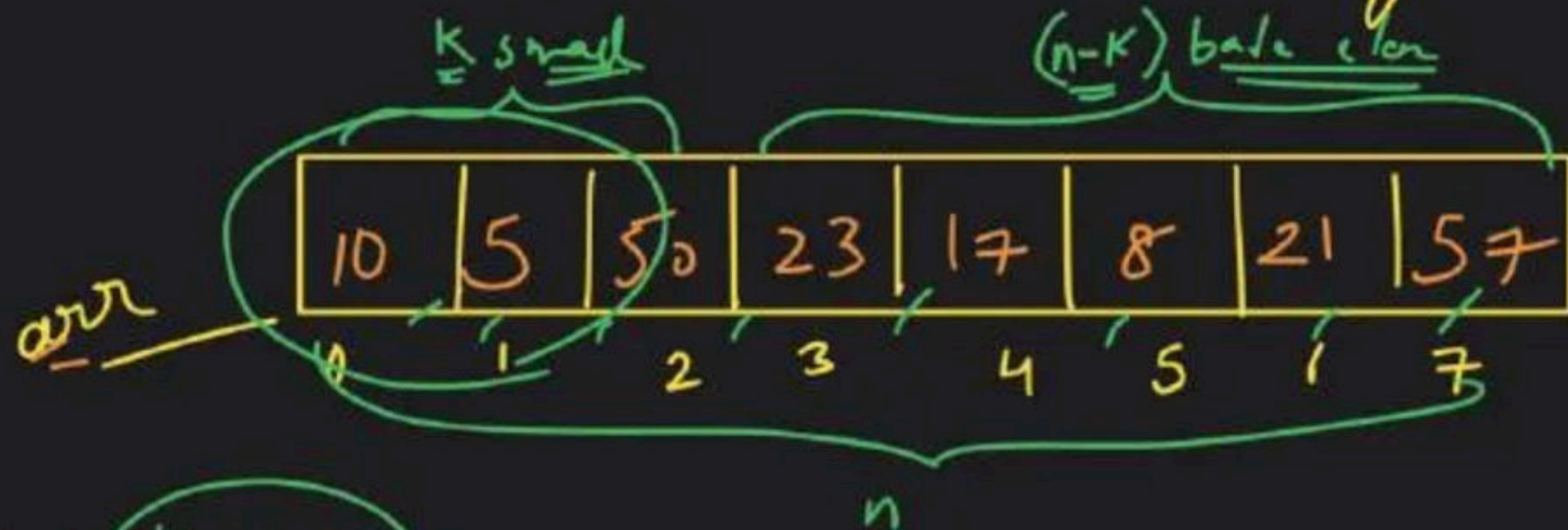


→ Map → Left
heapify X
heapsort



→ K^{th} Smallest element in an array

$K=3$



max-heap → K size → K small element → top

n size
 ↳ min heap

→ top → 1st small

↳ K pop → K^{th} smallest

→ ans ✓

Sort

↳ T.C → $n \log n$

Quick

↳ map

↳ ?

min heap

↳ T.C → ~~$O(n)$~~

↳ ~~$S.C. \rightarrow O(n)$~~

$O(n \log n)$

Max-heap

K-size

$K = 3$

K							
7	10	15	8	23	52	14	20
p	1	2	3	4	5	6	7

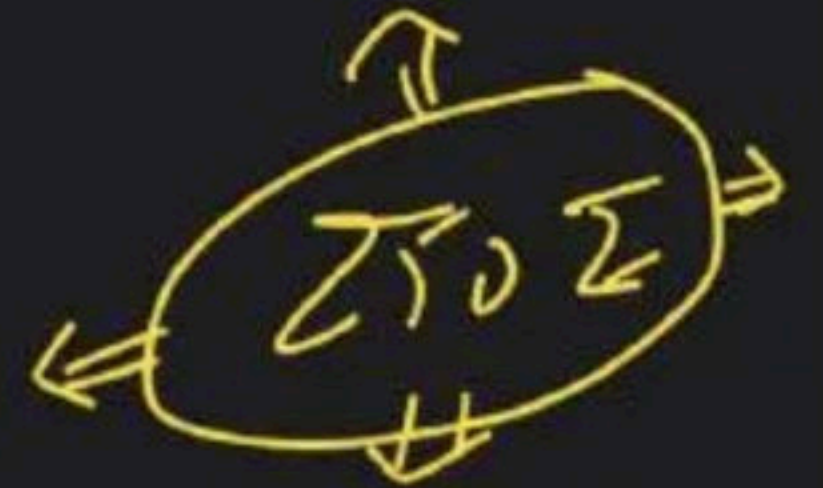


k^{th} largest elem

Sort $\rightarrow$ ✓

max-heap ✓

min-heap $\rightarrow$



3, 2, 3, 1, 2, 4, 5, 5, 6

↑ ↑ × × ↑ ↑

$k = 3$

→ kth largest

① k-size heap $\Rightarrow$ First k elements

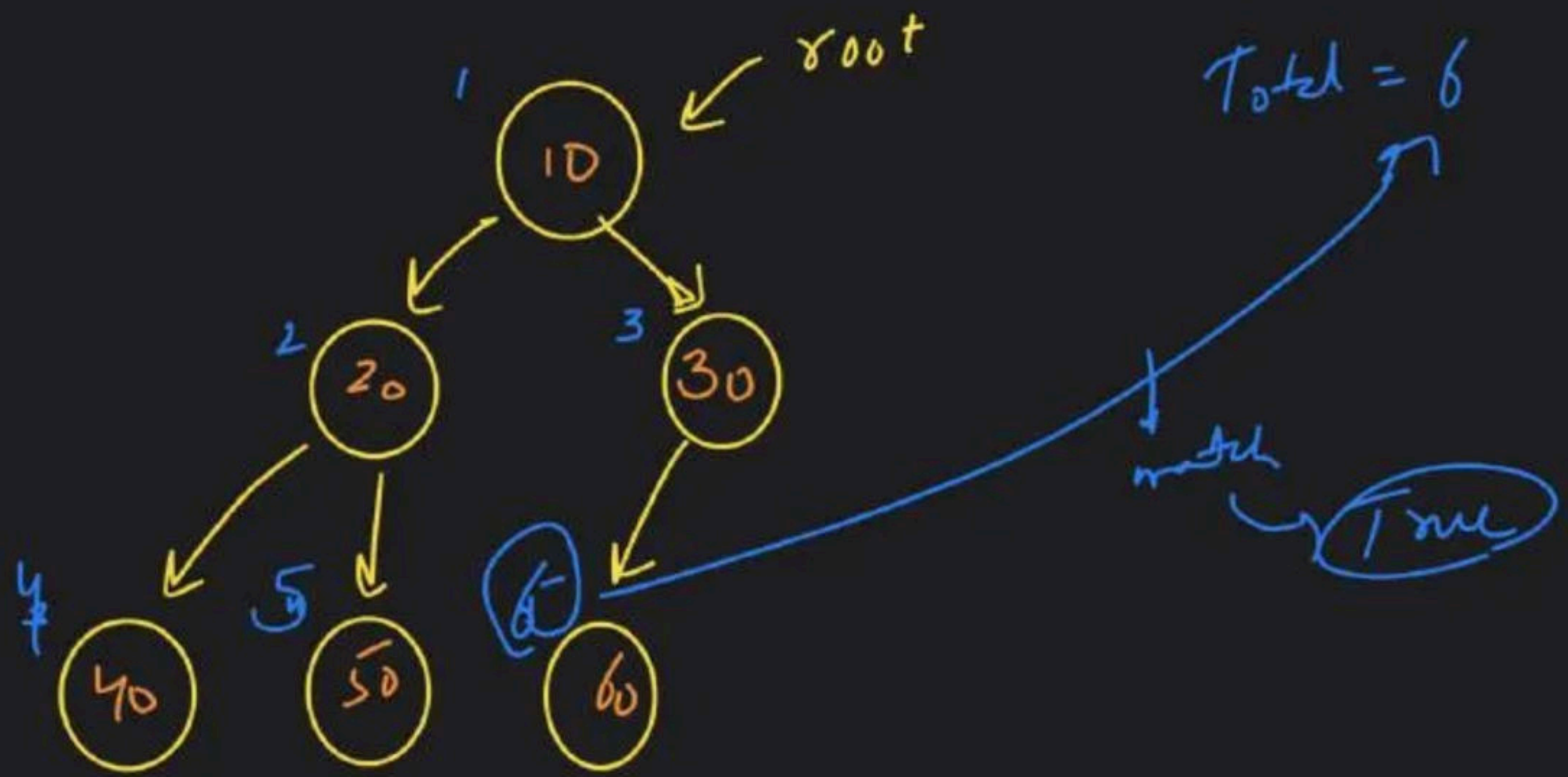
Min Heap

~~2~~
~~3~~
~~3~~
~~4~~
5, 5, 6

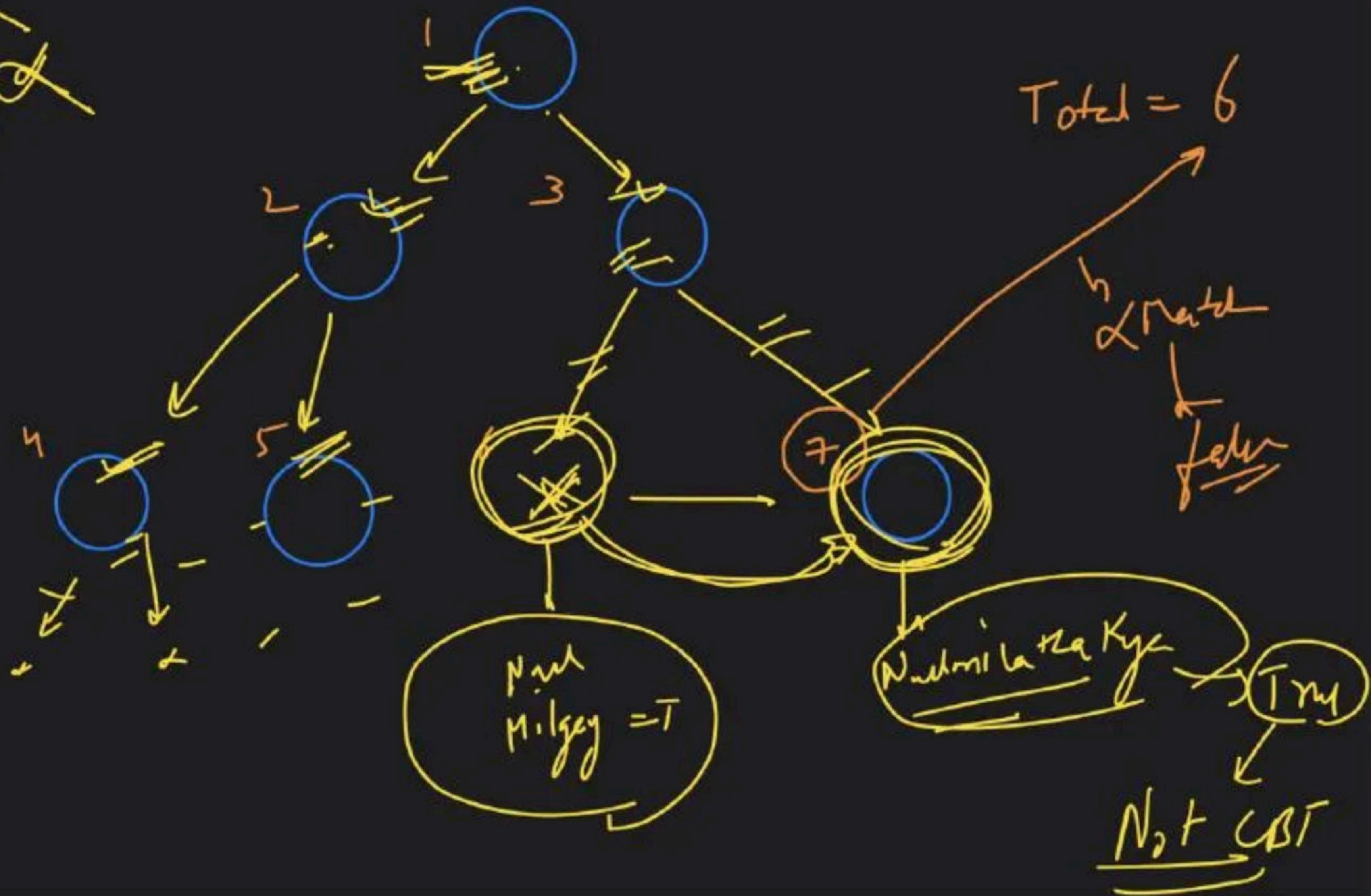


5

```
if (ele > Pa.top())
{
    Pa.pop()
    push el
}
```

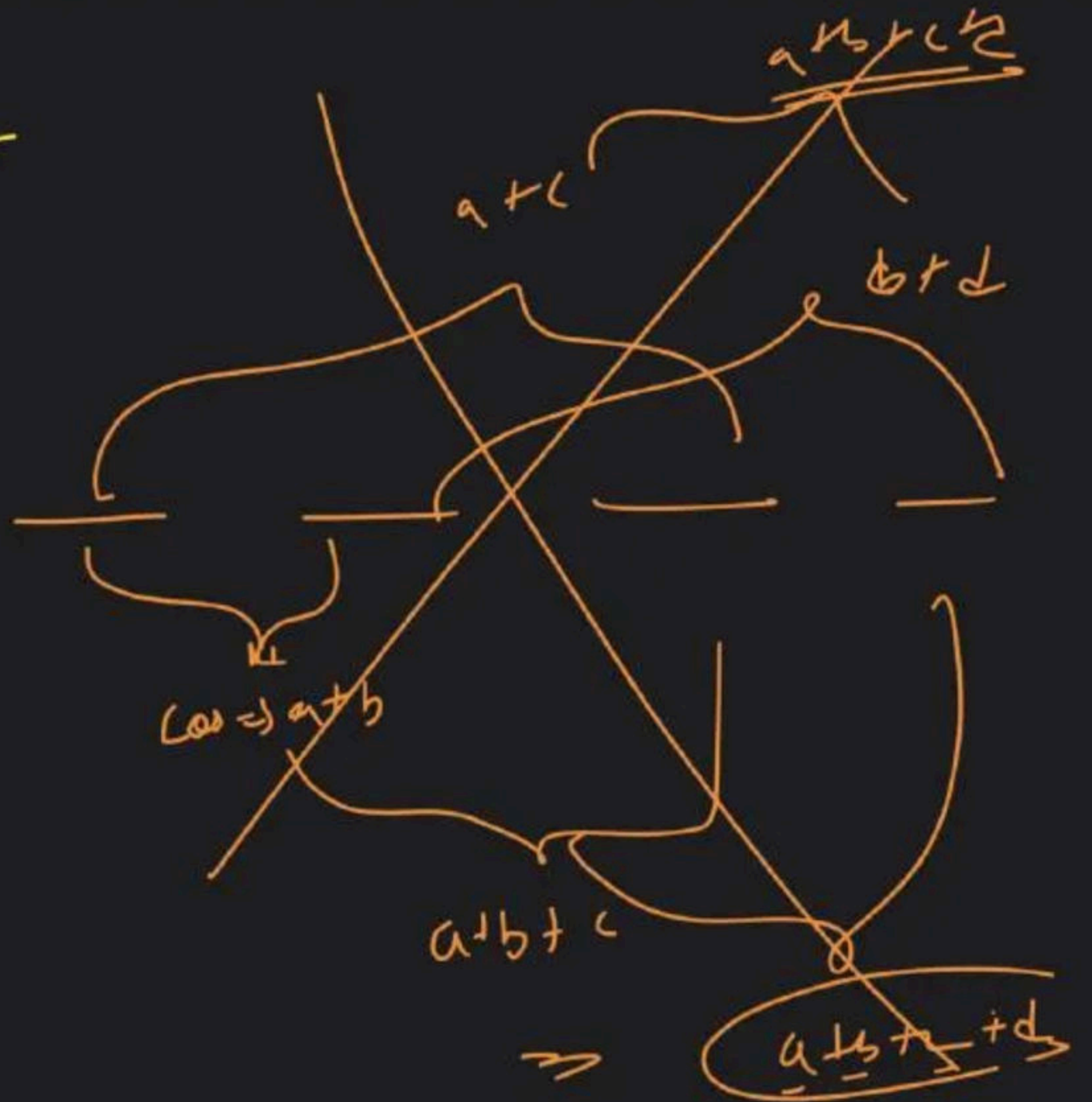



Count & Numbering



→ Expression tree → $4/3 \rightarrow ?$

→ Connect ropes



$$a + b + c + d$$

$$a + b + c + d$$

$$a + b + c + d$$

$$\begin{aligned} \text{Sum} &\downarrow \\ 0 &+ \\ a + b & \\ a + b + c & \\ a + b + c + d & \end{aligned}$$

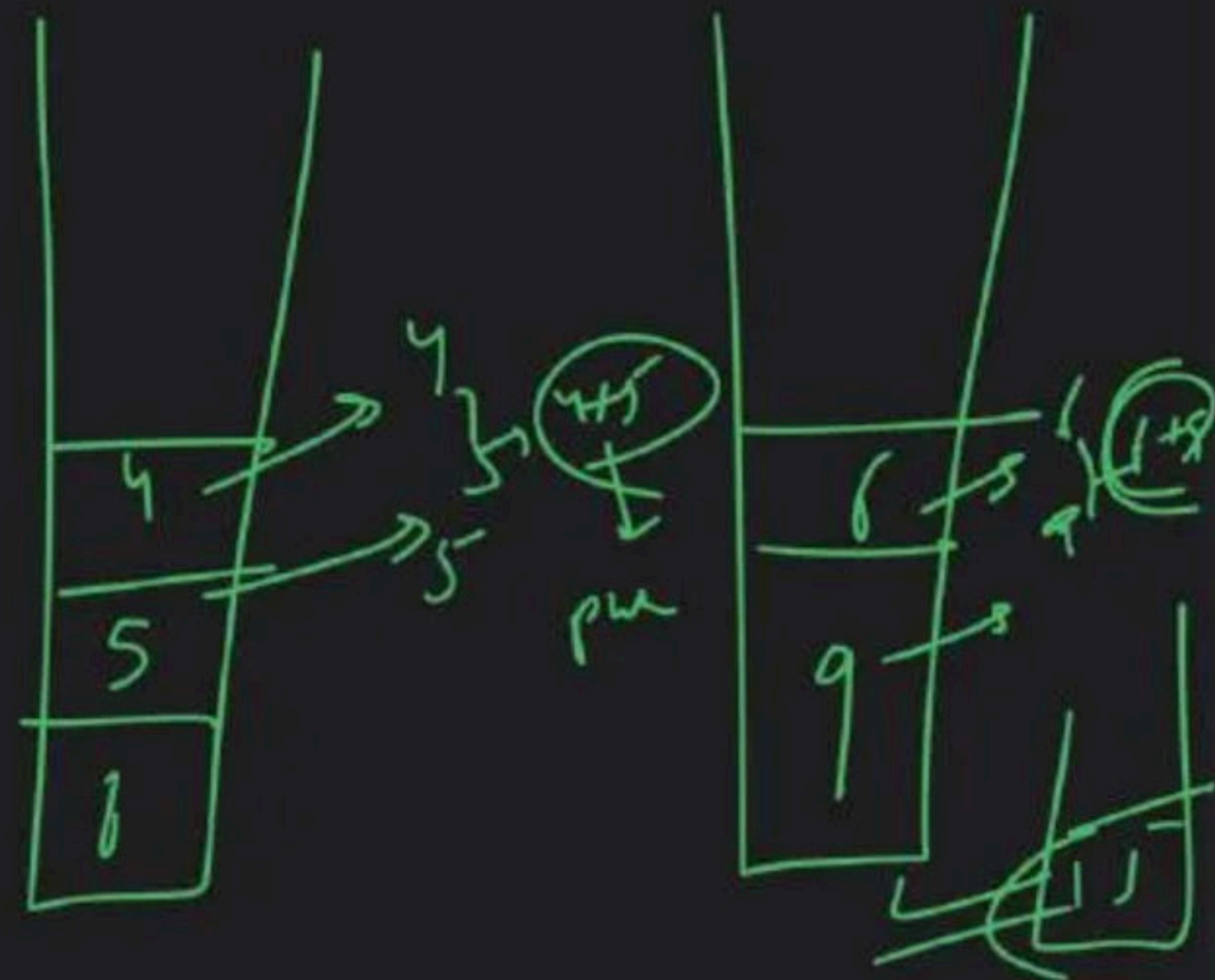
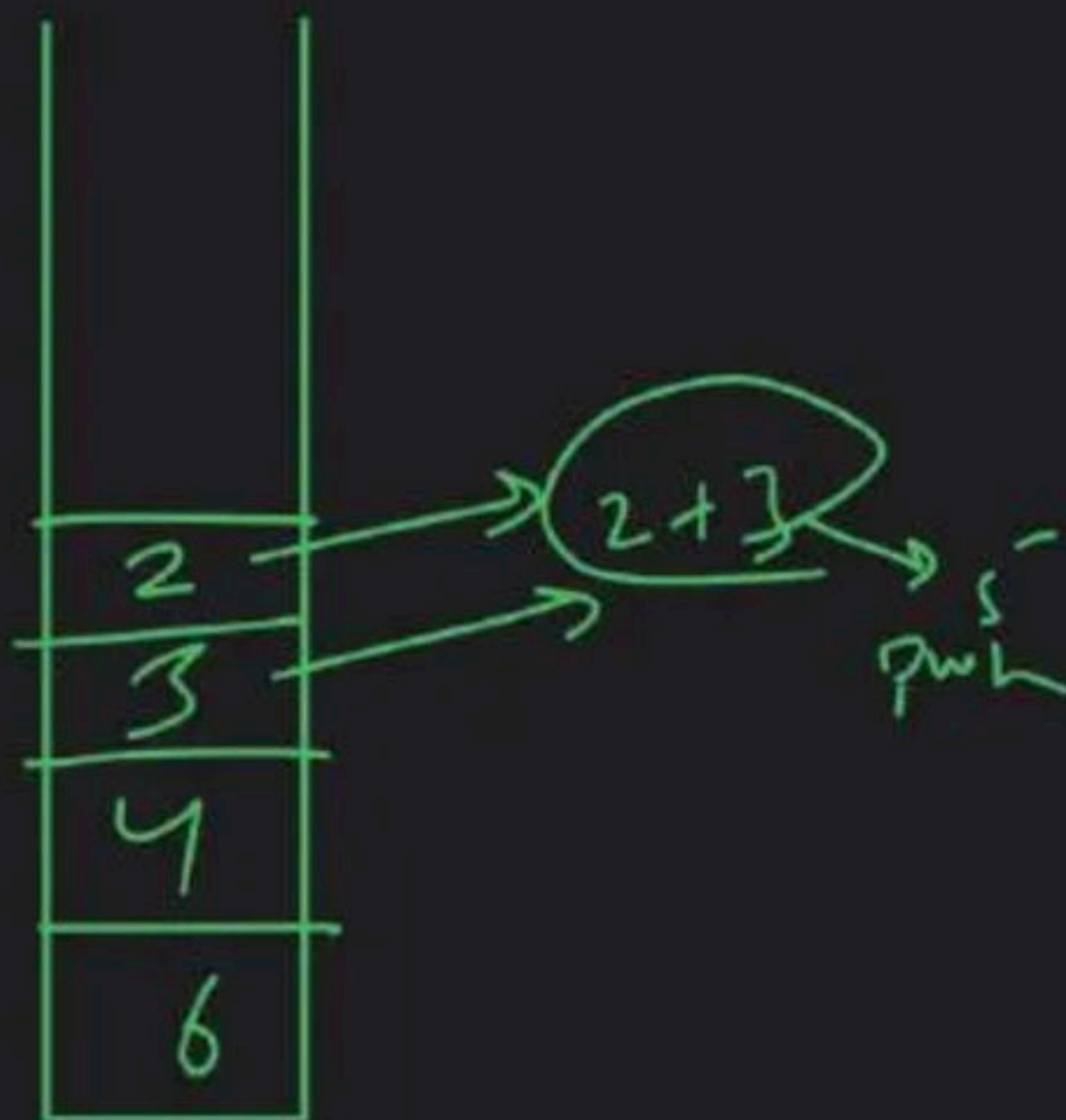
$$3c + 3b + 2c + d$$

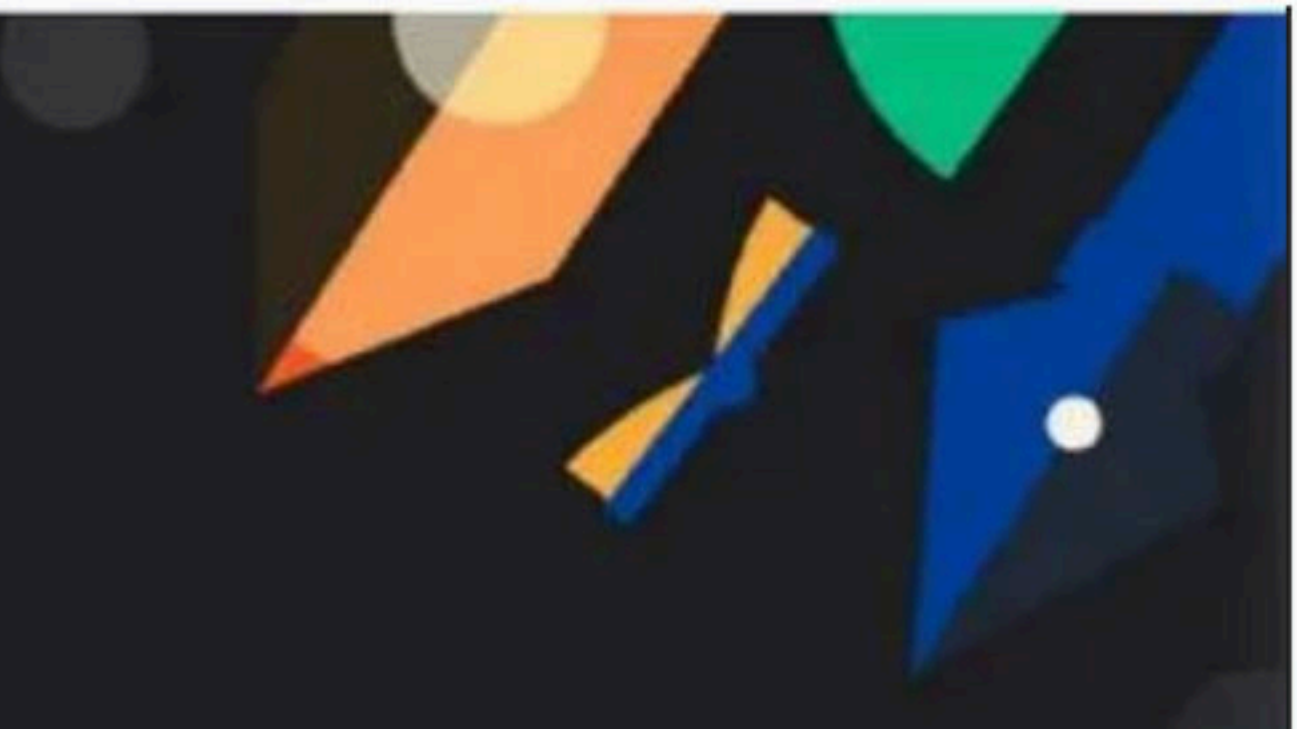
$$a + b + c + d$$

$$(2+5) \rightarrow \left. \begin{array}{c} 5 \\ 9 \\ 15 \end{array} \right\} \rightarrow (29)$$

4	3	2	6
---	---	---	---

min-heap

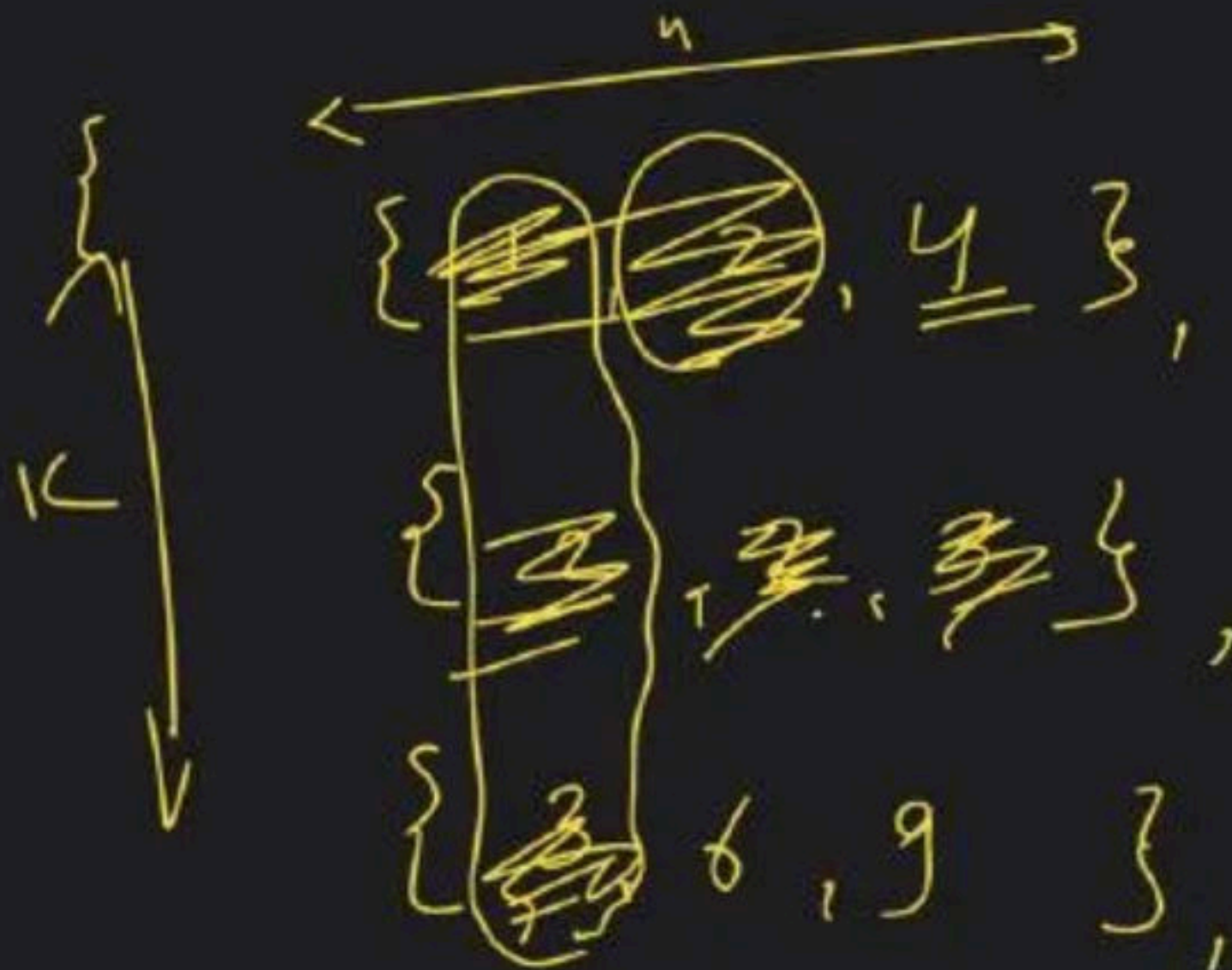




Heaps Class - 4

Special class

→ if -



sig way $\rightarrow$ 101
 $\hookrightarrow$ sort

$$T(n) = O(n \log(n))$$

set size

$\{2, 3\}$

$\{4, 3, 6\}$

$\{2, 2, 3\}$

$\{2, 6\}$

$\{4, 2, 3\}$

$\{4, 2, 3\}$

$\{4, 3, 3\}$

$\{6, 9\}$

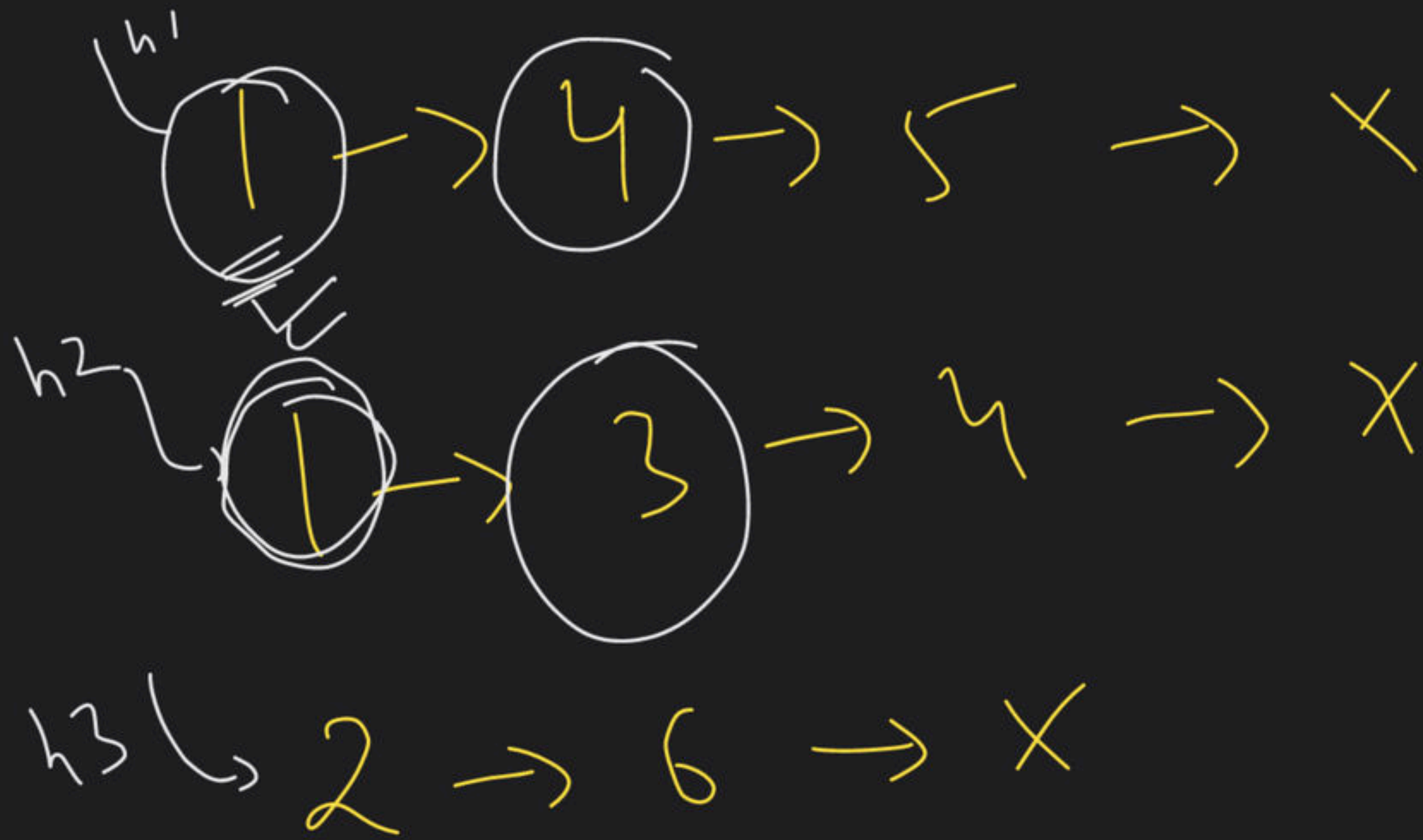
1	2	2	2	3	3	4	6	9
---	---	---	---	---	---	---	---	---

$$K = 3$$

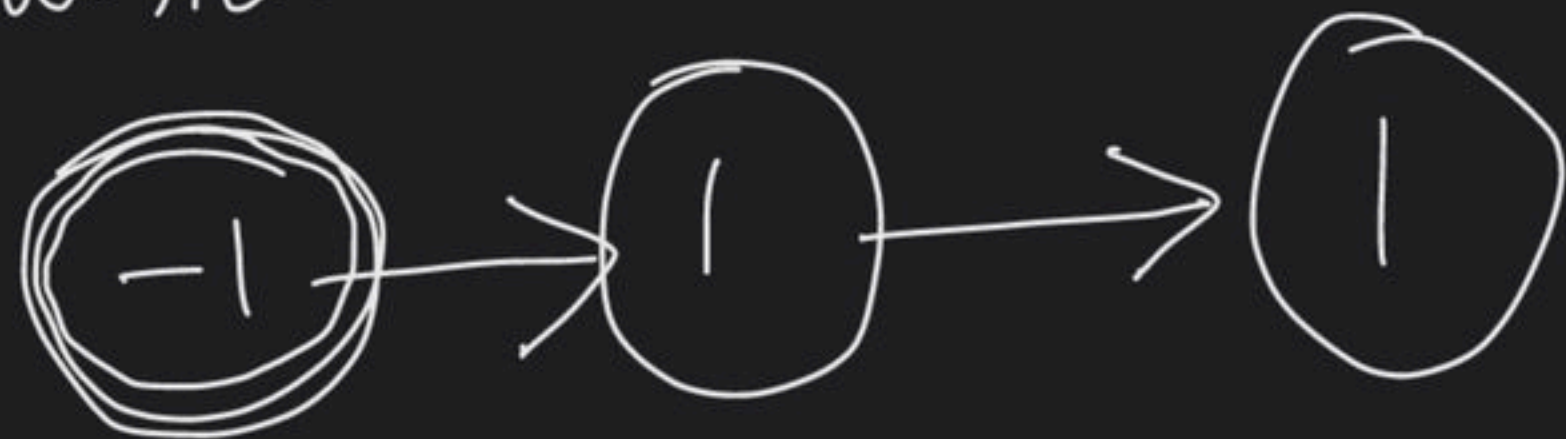
↳ 0 { 1, 2, 3 }

↳ 1 { 2, 4, 6, 8 }

↳ 2 { 6, 8, 9, 10, 11 }



ans, list

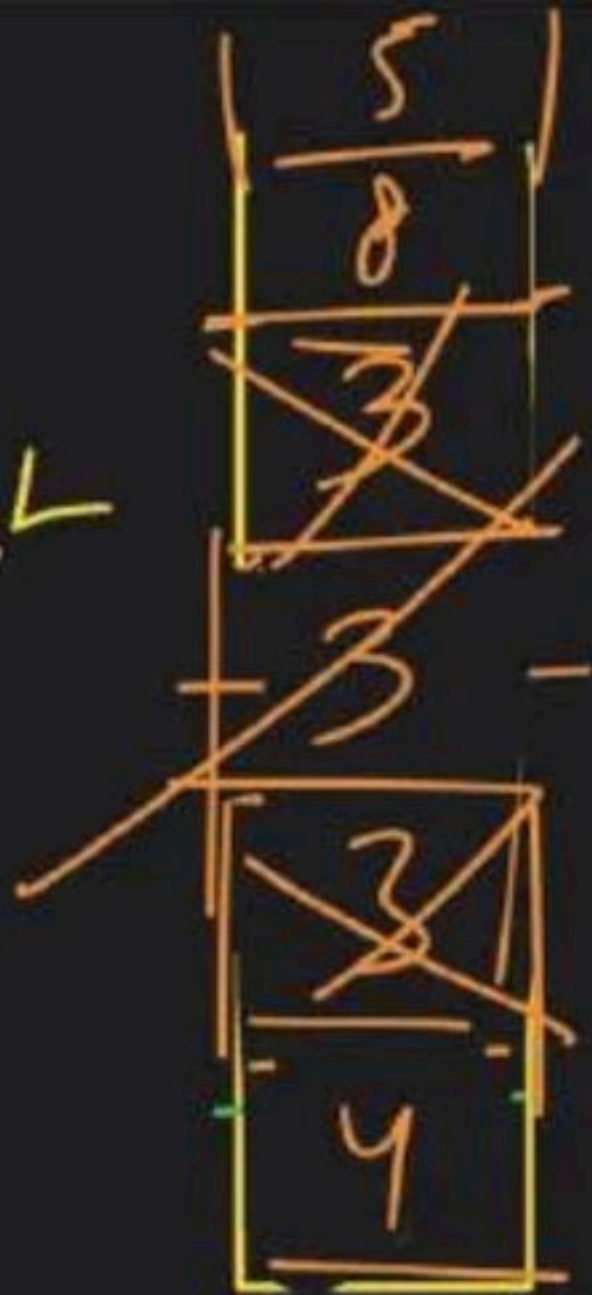


Min heap:



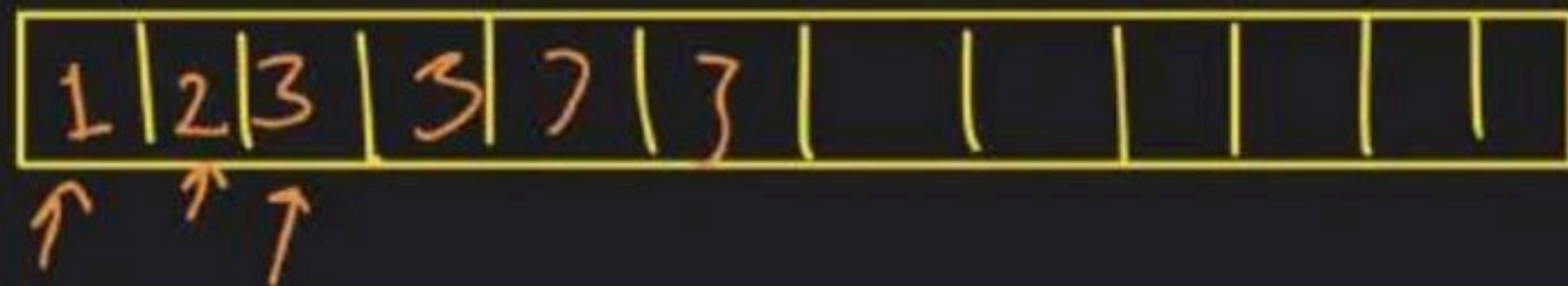
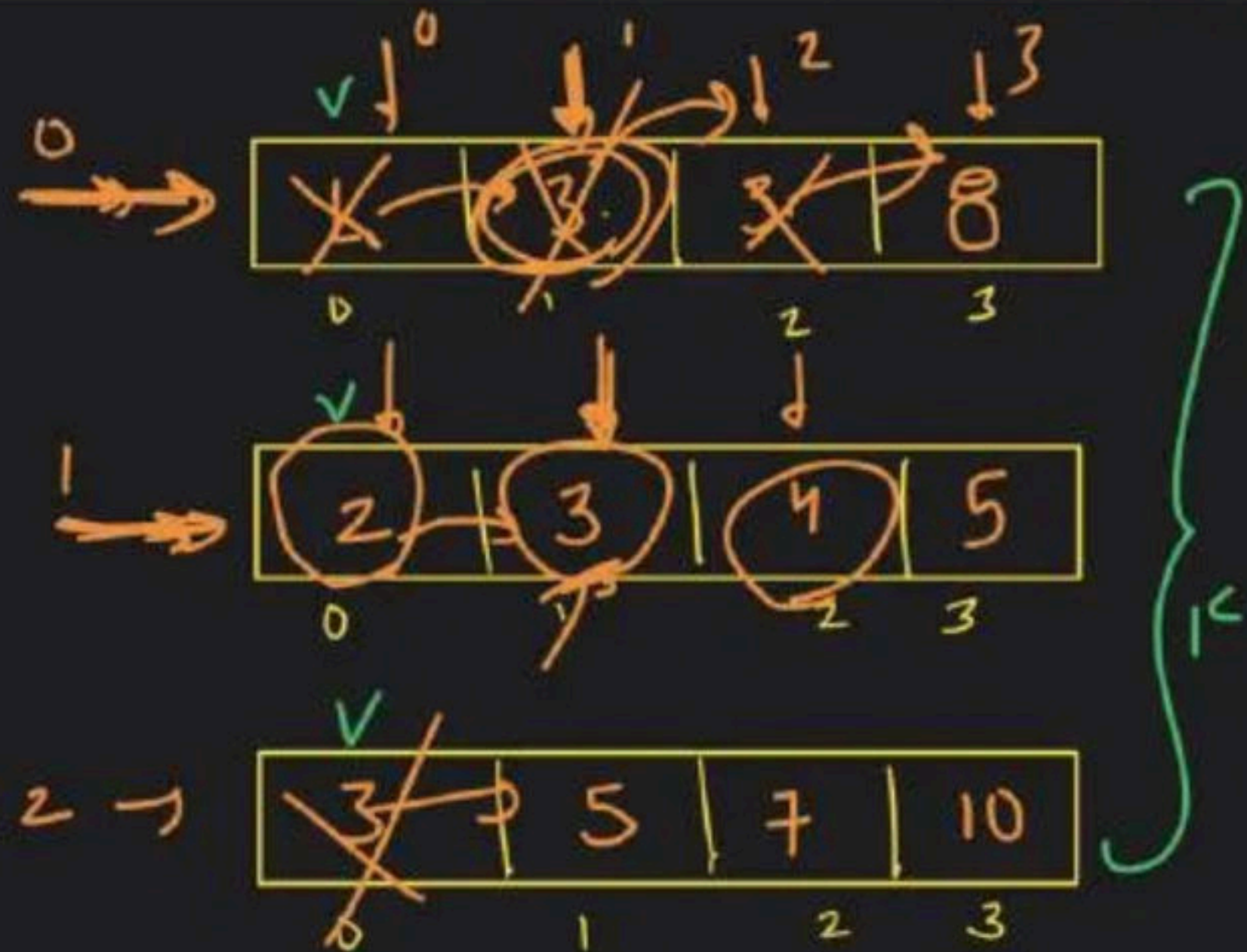
① create heap

② Fix k places
prev



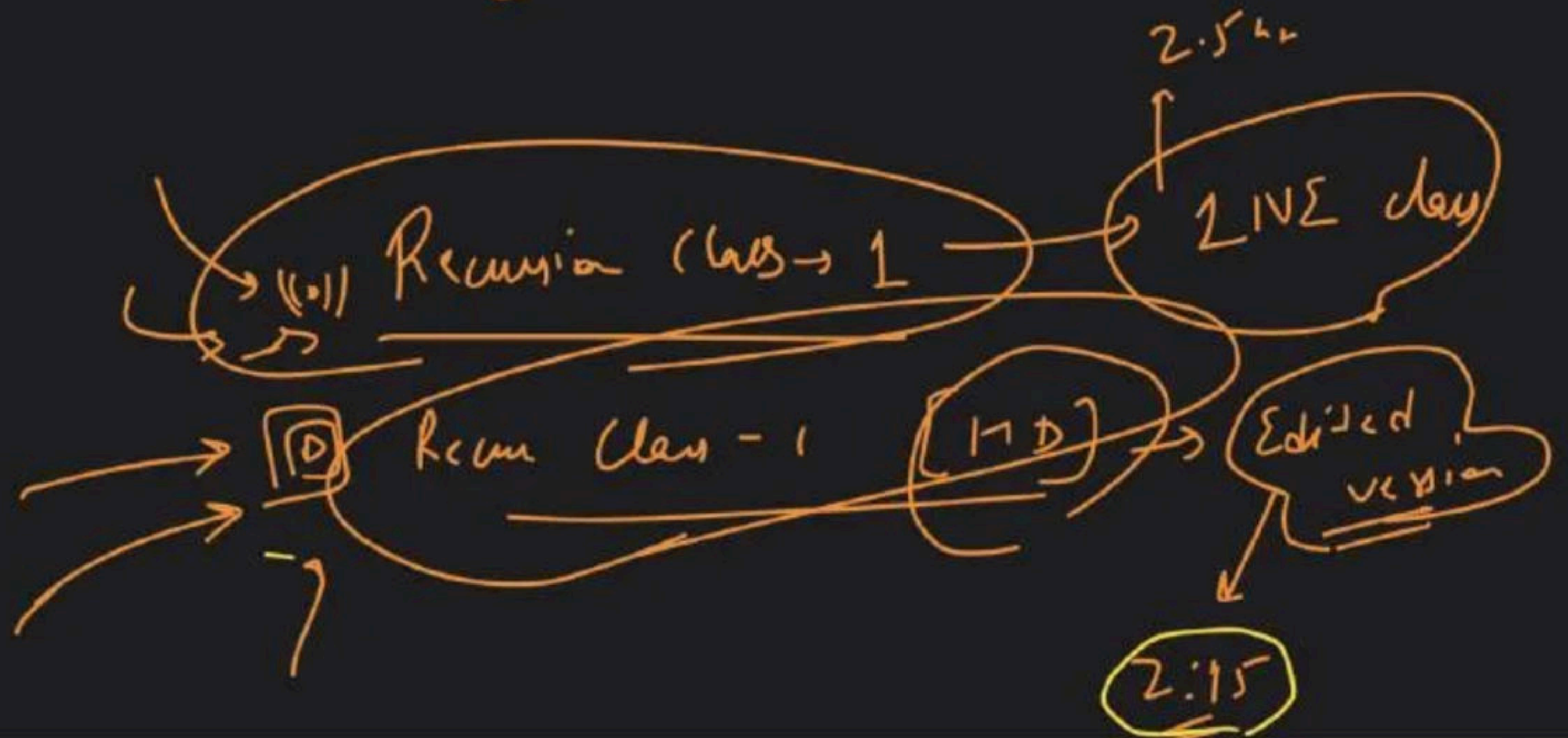
③ main Logic

- Front data
- ans insert
- after work
element check

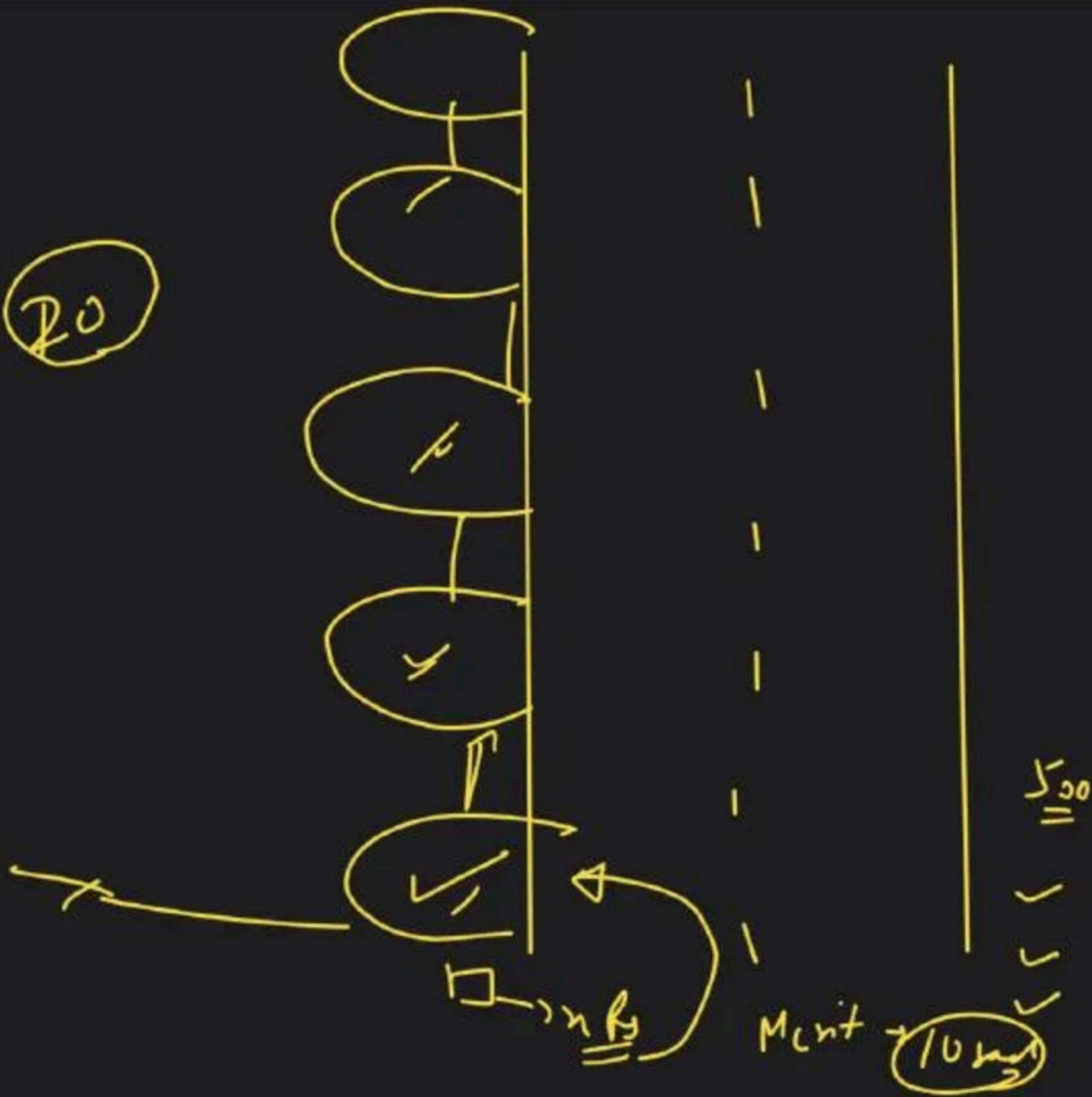


→ Merge K Sorted L-L

BST



20



FOMO

① create heap

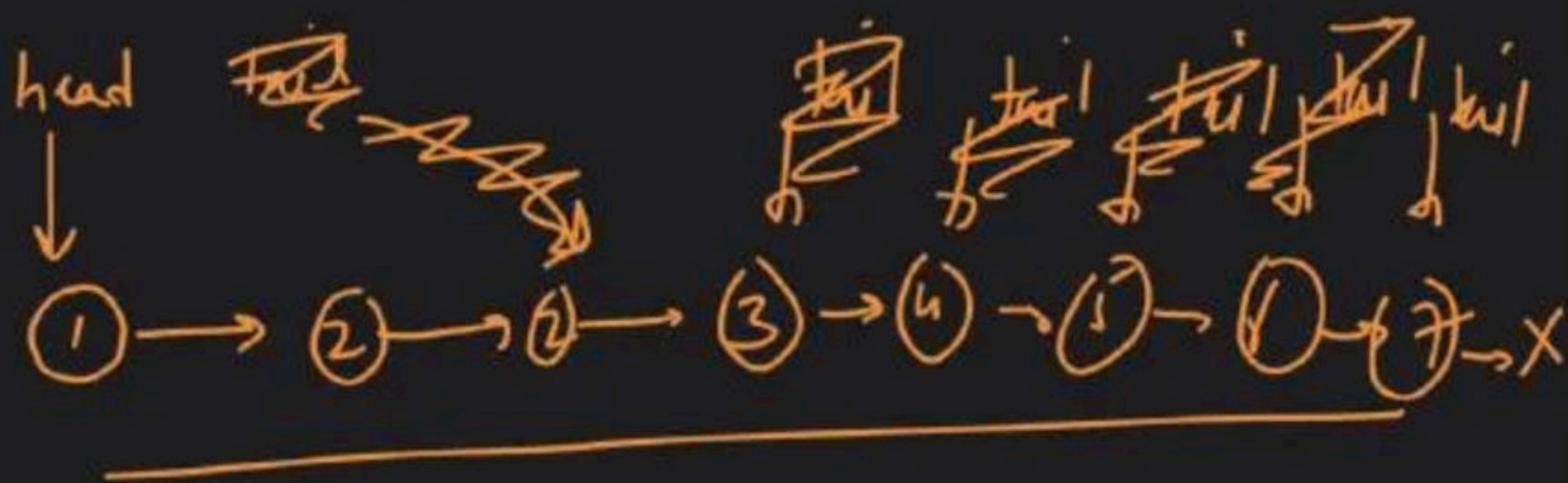
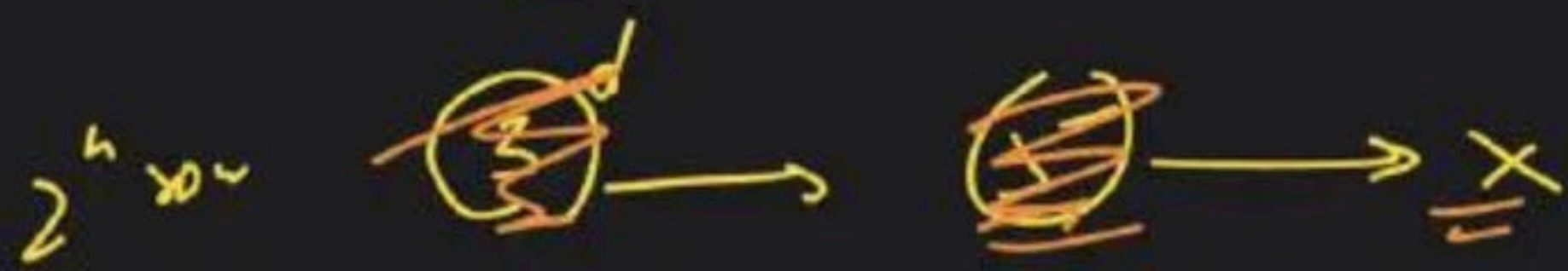
② K - element



③ main logic

↳ front
↳ ans store

↳ agar k element insert



→ Smallest Range in K List

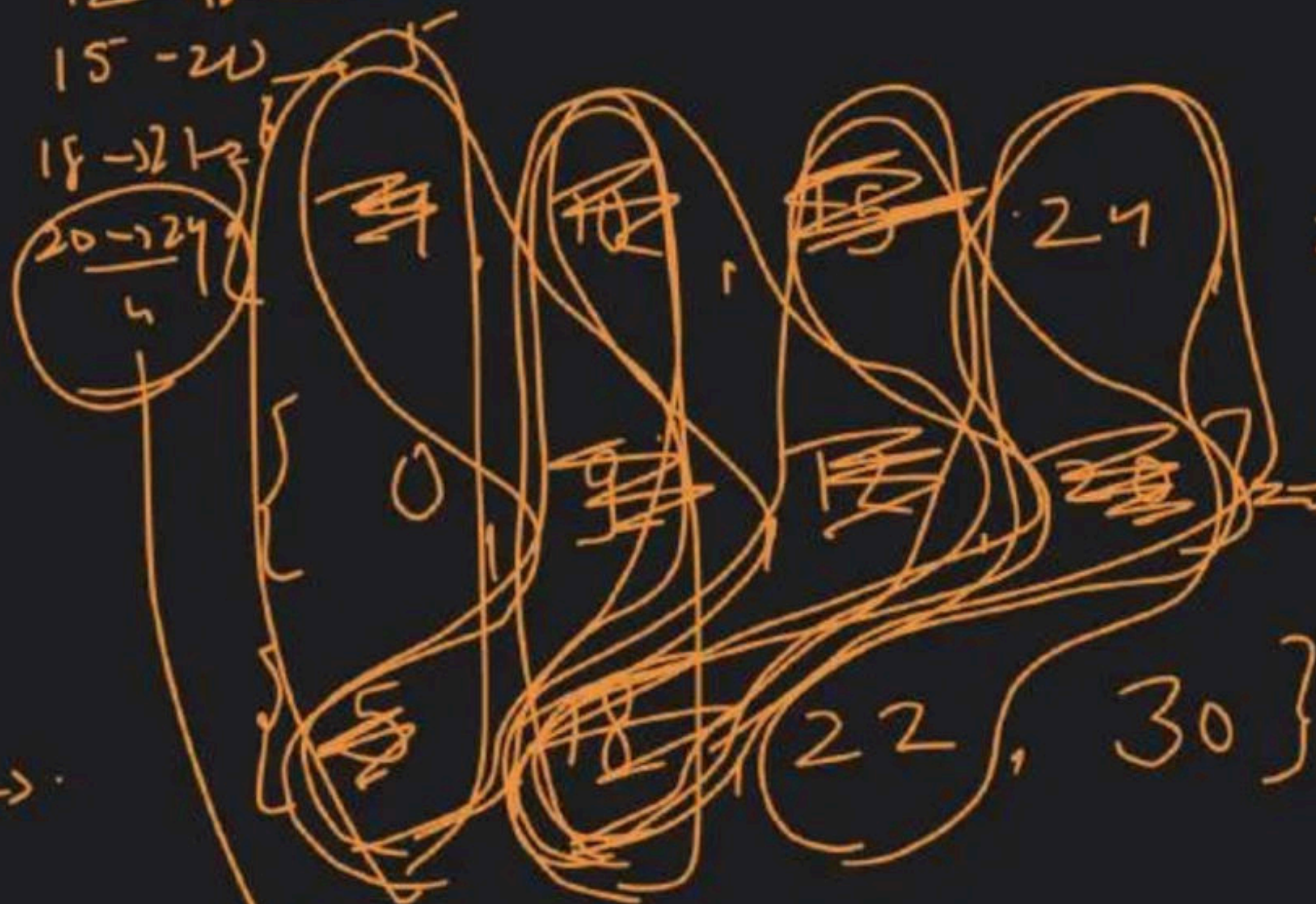
$\underline{5} \rightarrow 0 \rightarrow 5$
 $\underline{1} \rightarrow 4 \rightarrow 9$
 $\underline{5} \rightarrow 5 \rightarrow 16$
 $\underline{9} \rightarrow 9 \rightarrow 16$
 $\underline{5} \rightarrow 10 \rightarrow 18$

$\min \rightarrow 0$
 $\max \rightarrow 5$

~~5~~
 4
 5

4 →

$12 \rightarrow 18 \rightarrow 1$
 $15 \rightarrow 20 \rightarrow 5$
 $18 \rightarrow 21 \rightarrow 6$
 $20 \rightarrow 24 \rightarrow 4$



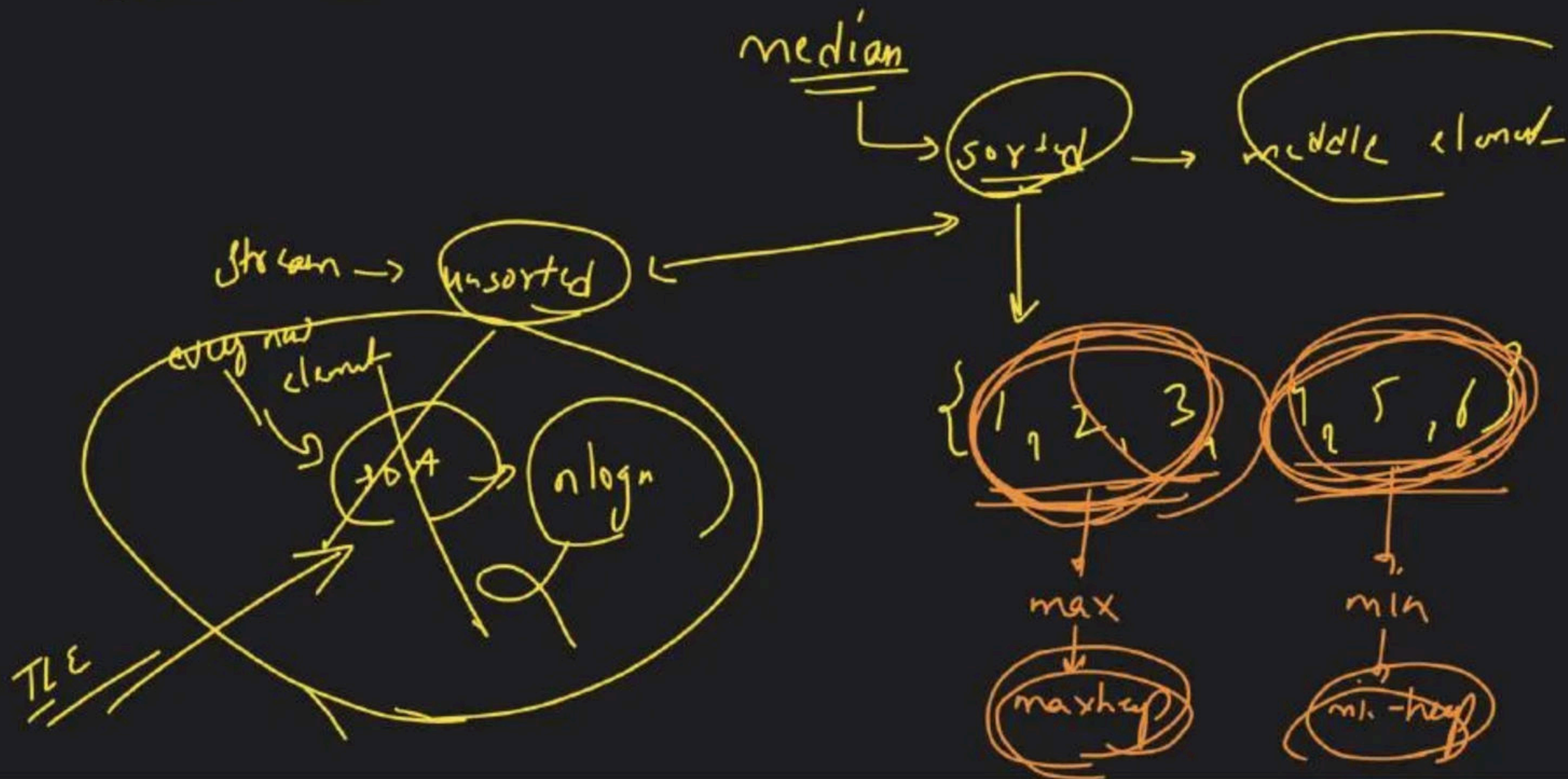
min range

$x \rightarrow y$
 $[x, y]$

$\swarrow$ suk gyl $\nearrow$ min → max



→ Median in a Stream



1, 2, 3, 4, 5, 6

N
4, 5

2 1 3 4 6 5

MaxHeap

MinHeap

1 2 3 4 5 6 7 8

Max heap

1, 2, 3, 4, 5, 6, 7

→ min

$7/2 \rightarrow 3.5$

3.5

Max

min

+

=

odd case

4 3

odd → Max heap has extra

even → Max heap element count
= Min "

① { 1 } ^{Max H}

[] ^{Min Heap}

2.

Max Heap, Min Heap

odd case

↳ find Median



MaxHeapTop

even case

↳

$\frac{\text{MaxTop} + \text{MinTop}}{2}$

2

Add Num
=

```
if maxHeapSize == 0 ||  
    num < maxHeap.top()  
    maxHeap.push(num)  
else  
    minHeap.push( )
```


0 { 1, 2, 3, ~~4~~ } { 4, 5, 6, 7 }

Man

↓
min

{ 1, 2, 3, 4, 5 }



Max

Size = 5



~~{ 1, 2, 3, 4, 5 }~~ { 6, 7, 8, 9 }

min
=

size = 4



Max
siz $\rightarrow$ Min
Heap

almost 1

Left
 $\downarrow$
Max
H

Right
 $\downarrow$
Min
Heap

$\rightarrow$ size $\Rightarrow$ size $\leftarrow \frac{Lp + up}{2} = \text{mid}$

$\rightarrow$ size $< 1 = \text{size}$

(1)
 Max Min → Median → 1

(1) (2)
 Max Min → Median = $\frac{1+2}{2} = 1.5$

(1) (2) (3) → Median → (2)
 Max Min
 (1) (2) (3) (4)
 Max Min → Median = $\frac{2+3}{2} = 2.5$

1
 { 1, 2 }
 { 1, 2, 3 }
 element
 ↳ Median
 ↳ Median
 ↳ Right
 ↳ Median
 ↳ Left

{ 1, 2, 3, 4 }

{ 1, 2, 3, 4, 5 }

{ 1, 2, 3, 4, 5, 6 }

→ Max size == Min size

element → Median → > Median → Right

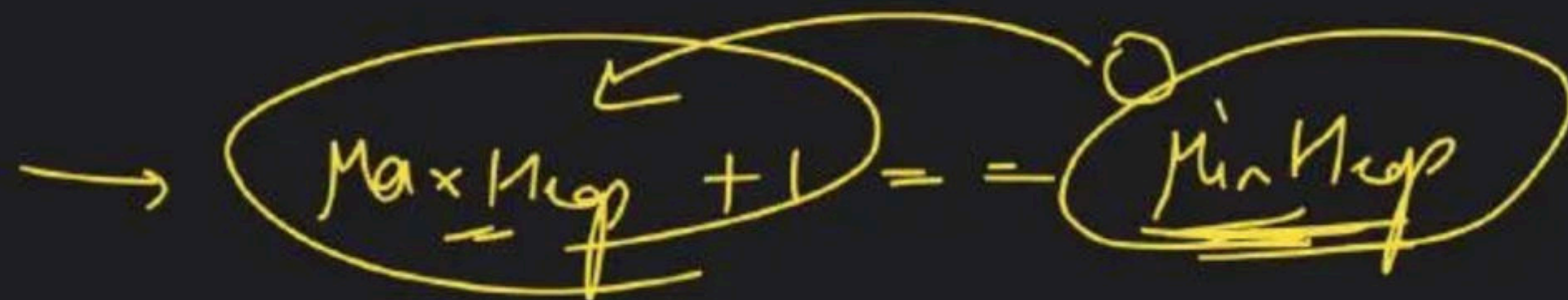
< Median → left

→ Max Heap == Min Heap size + 1

element → Median → > Right → straight

< left → max top → Min
→ element - 1 Max

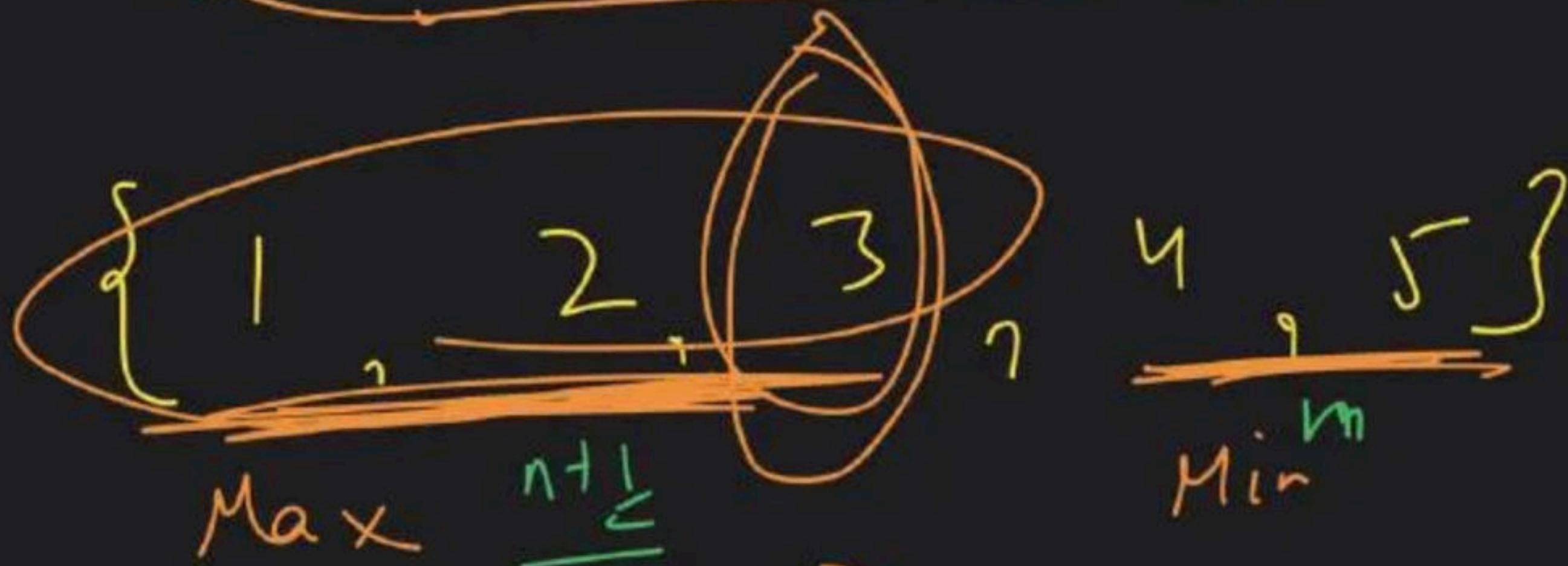
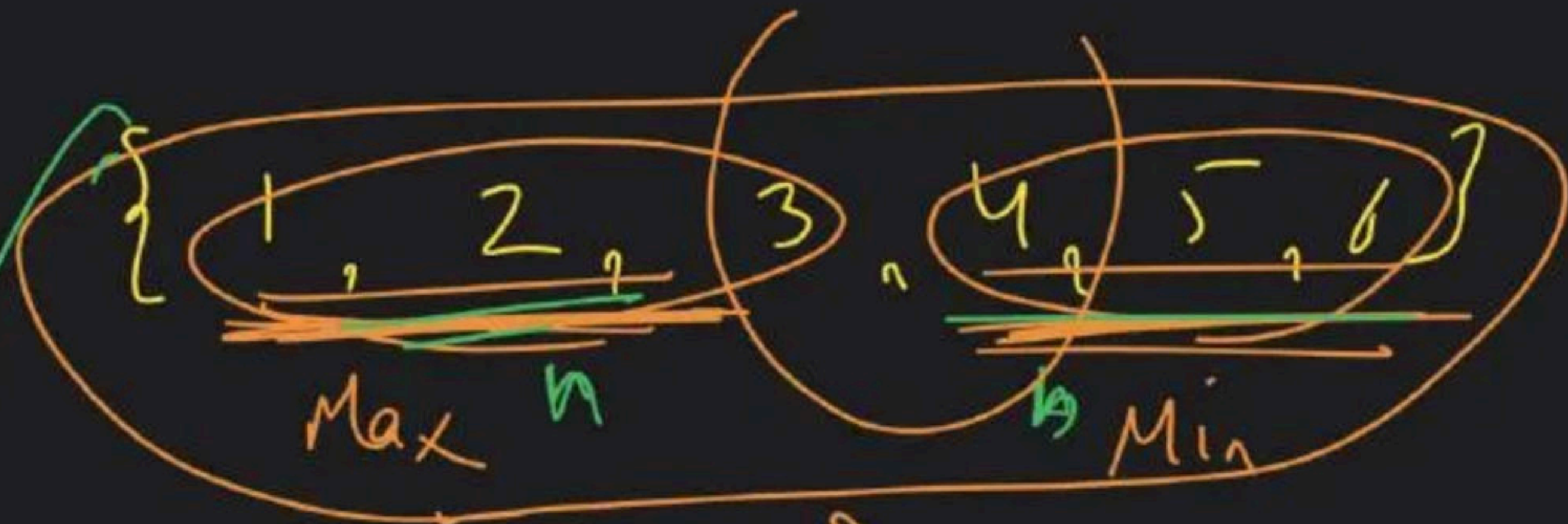
A



element > Median $\rightarrow$ $\rightarrow$ Right $\rightarrow$ Min heap $\rightarrow$ max
element $\rightarrow$ insert

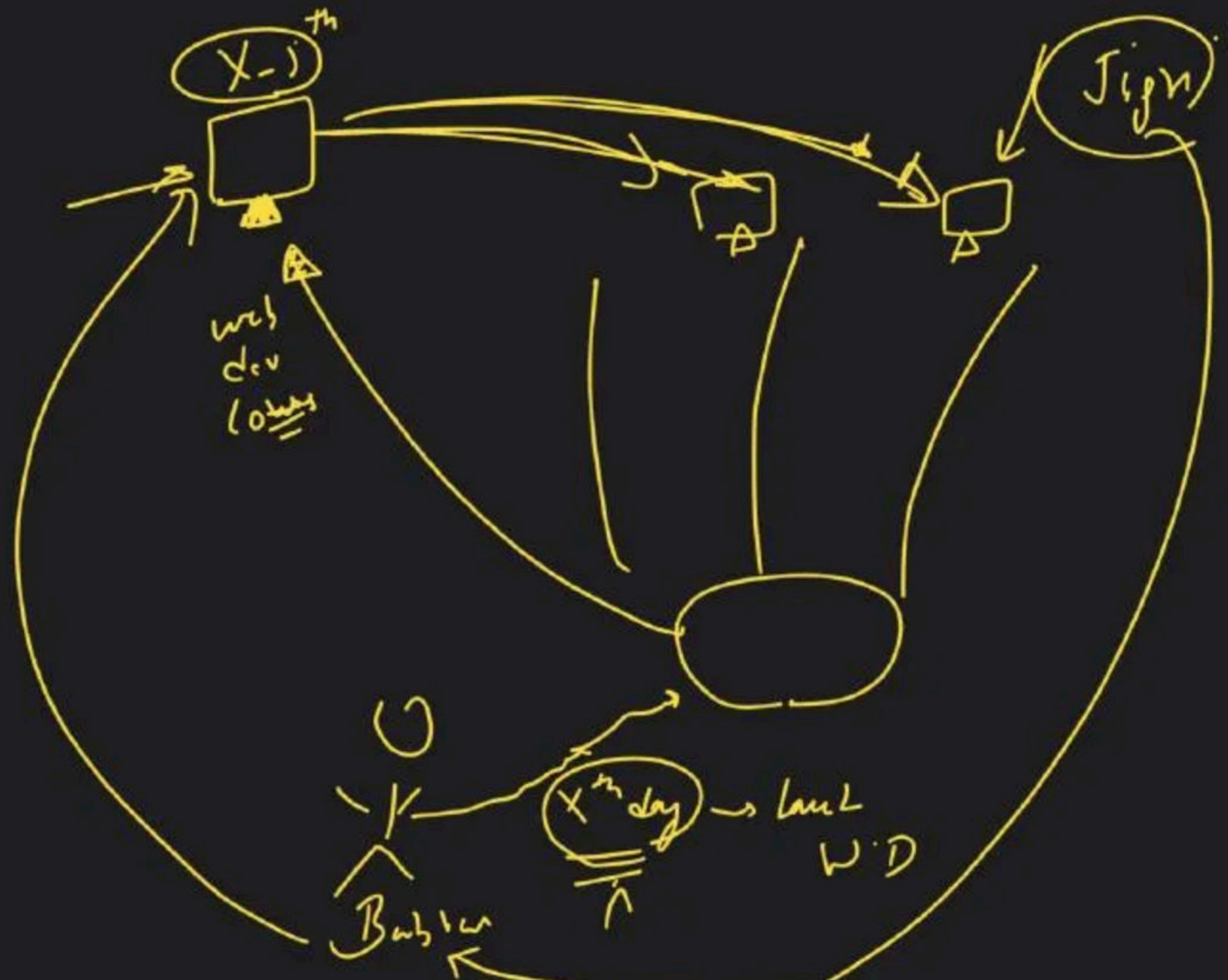
$\leftarrow$ Left $\rightarrow$ straight

3 cases



{ 1, 2, 3, 4, 5, 6, 7 }

Max Min



Kal

8:30

1 Map Quia

Q $\rightarrow$ Median in a Stream

